Mathematics-1 for Data Science

Week 1 Notes

Course: Mathematics–1

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Lecture 1 — Natural Numbers and Integers

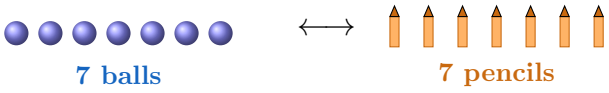
Counting, zero, $\mathbb{N}$ and $\mathbb{Z}$, arithmetic, divisibility, primes and prime factorisation

What this lecture covers

We begin at the very beginning: what a number is, why zero matters, and how the simple act of counting leads us step by step to the natural numbers and then to the integers. Along the way we look at the four arithmetic operations, exponentiation, quotients and remainders, divisibility, factors, prime numbers, and finally the prime factorisation of an integer.

1 Numbers and Counting

We use numbers, first of all, for **counting**. Suppose we see 7 balls, and separately we see 7 pencils. We need a way of saying that these two collections contain the same amount of things, and for that we use the number 7.



So 7 represents what is *common* to these two collections. In that sense 7 is an **abstract concept**: it refers to a quantity, not to any particular set of objects.

The most important number of all: zero

Perhaps the most important number of all, and one of Indian origin, is 0. It is essential to have a way of representing a quantity when there is *nothing* to count. Without a zero we could not use the place-value numbering system with which we write and manipulate numbers.

2 Natural Numbers

Natural numbers

The numbers used for counting, *starting from* 0, are the **natural numbers**:

$$\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$$

We write the set with a doubled letter $\mathbb{N}$.

There is some genuine confusion in the literature here. Many books use only $1, 2, 3, 4, \dots$ for the natural numbers and leave 0 out. To emphasise that 0 is included, one often writes a subscript:

$$\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}.$$

We shall write either $\mathbb{N}$ or $\mathbb{N}_0$, but in these notes, whenever we speak of natural numbers, **zero is always included**.

Which operations stay inside $\mathbb{N}$?

On natural numbers we can perform the usual arithmetic operations: addition, subtraction, multiplication and division. The interesting mathematical question is:

If we take natural numbers and perform an operation on them, do we always get back a natural number?

The first operation that fails this test is **subtraction**. If we subtract a larger number from a smaller one we fall below zero:

$$5 - 6 = -1 \notin \mathbb{N}.$$

If you have 5 things you simply cannot take away 6 things — and that is what subtraction means. So we must **extend the scope of our numbers** so that this operation works sensibly. This is how the negative numbers arise.

3 Integers

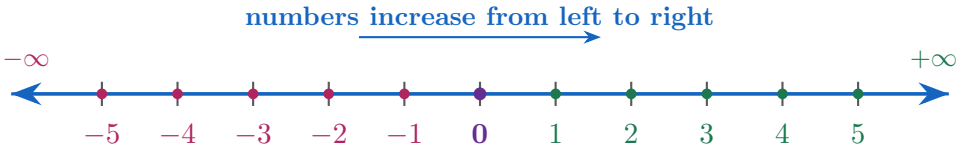
We started with the counting numbers $1, 2, 3, 4, \ldots$; we added 0 to account for counting nothing (note that 0 is neither positive nor negative, so $0, 1, 2, 3, \ldots$ are technically the *non-negative* numbers); and now we add, symmetrically on the other side, the negative numbers $-1, -2, -3, \ldots$

Integers

The set of natural numbers extended by the negative numbers is the set of **integers**:

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$$

Both $\mathbb{N}$ and $\mathbb{Z}$ are infinite sets, but they are infinite in different ways: $\mathbb{N}$ has a **starting point** at 0 and runs forward, whereas $\mathbb{Z}$ extends to infinity in **both** directions.

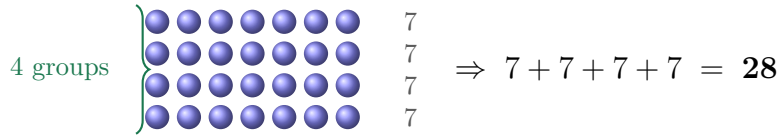


It is very convenient to picture the integers arranged in this way, with the small numbers on the left and the large numbers on the right. This picture is called the **number line**.

4 Multiplication as Repeated Addition

Subtraction took us out of $\mathbb{N}$ and forced us to invent $\mathbb{Z}$. Let us now examine the remaining two operations, multiplication and division.

When we say 7×4, what we really mean is: take 7 objects and make 4 copies of them.



$$\Rightarrow\ 7 + 7 + 7 + 7 = \mathbf{28}$$

$$7 \times 4 = 7 + 7 + 7 + 7 = 28 \qquad \text{and in general} \qquad m \times n = \underbrace{m + m + \cdots + m}_{n \text{ times}}$$

That is, multiplying m by n means making n copies of m. **Multiplication is repeated addition.**

Notation for multiplication

The symbol $\times$ is cumbersome when we write out equations, so we often replace it by a dot, and sometimes we write nothing at all:

$$7 \times 4 \quad \equiv \quad 7 \cdot 4 \qquad\qquad m \times n \quad \equiv \quad m \cdot n \quad \equiv \quad mn$$

Why we still write the dot between numerals

For actual numerals we normally keep the dot. If we were to write 7 4 side by side, you would not know whether it means the number 74 or the product 7×4. So for numbers we write $7 \cdot 4$ explicitly. But when letters such as m and n stand for numbers, writing mn unambiguously means one number m multiplied by another number n.

The rule of signs

Integers carry signs, so when we multiply we must also determine the sign of the result. Let m be a positive number, so that $-m$ is negative.

$$(-m) \times n = -(mn) \qquad (-m) \times (-n) = +mn$$
$$(-7) \times 4 = -28 \qquad (-7) \times (-4) = +28$$

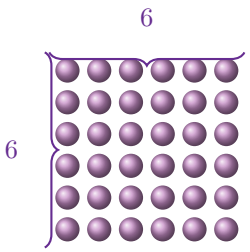
In the second case the two negations cancel. In general:

even number of minus signs $\Rightarrow$ result is **positive**

odd number of minus signs $\Rightarrow$ result is **negative**

5 Exponentiation as Repeated Multiplication

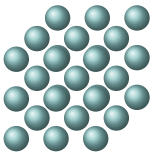
Just as we obtained multiplication from repeated addition, we can repeat multiplication. Instead of $m + m$ we take $m \times m$, which we call m **squared**.



$$6 \times 6 = 6^2 = 36$$

The name *square* is visible in the picture: 6×6 balls can be arranged in a square. The notation m^2 records the fact that m is multiplied by itself twice.

If we multiply m by itself three times we get a **cube**. Here 3 balls by 3 balls make a square of 9 balls, and we stack 3 such layers one on top of the other, which naturally forms a cube.



$$3 \times 3 \times 3 = 3^3 = 27$$

$$m^2 = m \cdot m, \qquad m^3 = m \cdot m \cdot m, \qquad m^k = \underbrace{m \cdot m \cdots m}_{k \text{ times}}$$

Why the names stop at "cube"

We live in a three-dimensional world and cannot picture an object with more than three dimensions, so our everyday vocabulary stops at the cube. For a general exponent we have no fancy name: we simply read m^k as *the k-th power of m*.

The ladder of operations

Multiplication is repeated addition

$$\Longrightarrow$$

Exponentiation is repeated multiplication

6 Division, Quotient and Remainder

Here is the familiar school problem. You have a certain number of objects and you wish to divide them among a certain number of people.

Example — 20 mangoes among 5 friends

You have 20 mangoes and 5 friends. How many mangoes does each friend get?
Start by distributing one mango to each friend. That uses up 5 mangoes, so 15 remain. Repeat the process: 15 becomes 10, then 10 becomes 5, then 5 becomes 0. After 4 rounds — each round giving away 5 mangoes — all the mangoes are gone.

$$20 - 5 - 5 - 5 - 5 = 0 \qquad \Longrightarrow \qquad 20 \div 5 = 4$$



4 rounds of subtraction ⇒ quotient = 4, remainder = 0

So **division is repeated subtraction**: we keep subtracting the divisor, and if we finally land exactly on 0, the division is exact.

Example — 19 mangoes among 5 friends

Now suppose there are only 19 mangoes. Distributing as before:

$$19 \xrightarrow{-5} 14 \xrightarrow{-5} 9 \xrightarrow{-5} 4$$

After three rounds only 4 mangoes are left, and there are 5 friends, so we cannot give one to each. We managed to distribute 3 times and 4 are left over.

$$\textbf{quotient} = 3, \qquad \textbf{remainder} = 4, \qquad 19 = 5 \times 3 + 4$$

Quotient, remainder and the modulus notation

The **quotient** is the number of times the divisor can be subtracted without going into a fractional part; whatever is left over and cannot be subtracted one more time is the **remainder**. *Modulus* is another word for remainder, written `mod`:

$$a \bmod b = \text{the remainder when } a \text{ is divided by } b$$

$$\boxed{19 \bmod 5 = 4}$$

7 Divisibility and Factors

With the modulus notation in hand we can define what a factor is. A factor is a number which divides a bigger number *evenly*, without any remainder.

Divisibility

$$a \mid b \iff b \bmod a = 0 \iff b = a \cdot k \text{ for some } k$$

We read $a \mid b$ as "a divides b". On the left of the bar sits the smaller number, on the right the bigger one. Equivalently, b is a **multiple** of a. If $a \mid b$ we also say that a is a **factor** of b.

Examples of divisibility

$$4 \mid 20 \quad \text{because } 4 \times 5 = 20$$

$$7 \mid 63 \quad \text{because } 7 \times 9 = 63$$

$$32 \mid 1024 \quad \text{because } 32 \times 32 = 1024$$

The symbol for *not* being a divisor is the same bar with a stroke across it:

$$4 \nmid 19 \quad \text{— there is no number we can multiply by 4 to get 19}$$

$$9 \nmid 100 \quad \text{— since } 9 \times 11 = 99 \text{ and } 9 \times 12 = 108$$

Factors come in pairs

It is easy to see that factors must come in pairs: if $a \mid b$, then a goes into b some k times, so $k \cdot a = b$, and therefore k is a factor of b as well.

Factors of 12

$$1 \times 12, \qquad 2 \times 6, \qquad 3 \times 4$$

$$\boxed{1 \times 12 = 12} \qquad \boxed{2 \times 6 = 12} \qquad \boxed{3 \times 4 = 12}$$

Note that 1 is a factor of every number, because $1 \times n = n$; so the partner of 1 is always the number itself.

Factors of 36 — the perfect-square case

There is an important side condition: sometimes a factor is paired with itself. This happens exactly when the number is a **perfect square**. Since $36 = 6 \times 6$:

$$1 \times 36, \qquad 2 \times 18, \qquad 3 \times 12, \qquad 4 \times 9, \qquad 6 \times 6$$

Here 6 does not produce a new partner — its partner is itself.

Counting the factors

- If a number is **not** a perfect square, its factors pair up completely, so it has an **even** number of factors: $2 + 2 + 2 + \cdots$
- If a number **is** a perfect square, all factors pair up except the square root, which appears only once, so it has an **odd** number of factors: $2 + 2 + \cdots + 1$

8 Prime Numbers

Prime number

A **prime number** is one which has no factors other than 1 and itself. We usually write p for a prime. Thus p has exactly **two** factors:

$$1 \quad \text{and} \quad p.$$

It is essential that there be two *separate* factors.

Why 1 is not a prime

Technically 1 is not a prime, because it has only *one* factor. Since $1 \times 1 = 1$, the only factor 1 has is 1 itself, and so it does not have two distinct factors.
Hence the smallest prime is **2**: it has the two factors 1 and 2, and no others. Likewise 3 is prime, since its only factors are 1 and 3 (2 does not go into 3).

The first few primes are

$$2,\ 3,\ 5,\ 7,\ 11,\ \ldots$$

Notice that after 2 no even number can be prime, since every even number is a multiple of 2 and so 2 divides it. Coming to 9: it is *not* prime, because it is a multiple of 3. But 11 is prime, and so on.

The Sieve of Eratosthenes

There is a clever way of generating primes, called the **Sieve of Eratosthenes**: whenever you discover a prime, you knock off all the numbers which are multiples of it.

The procedure

1. Lay out all the numbers up to n in a grid. We know 1 is not a prime, so leave it out.
2. Take the first number that has not been marked off; call it a prime.
3. Mark off all its multiples to the right as non-primes.
4. The next unmarked number is the next prime. Repeat.

Let us do this for the numbers from 1 to 100. We start with 2, our first prime, and knock off all multiples of 2 — that is, all the even numbers, which can be struck out a whole column at a time. The first number not yet marked off is 3, so 3 is a prime, and we now strike off the multiples of 3. Some of them, such as 6 and 12, are already gone because they are multiples of 2; but 9 and 15 are newly struck out. Continuing in this way, the next surviving number is 5, which is therefore prime, and so on.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

green = prime | **pink** = struck off (composite) | **grey** = 1, not a prime

A remark on efficiency

The sieve is not necessarily an *efficient* way to find prime numbers, but it is a good way to generate them without missing any.

9 Prime Factorisation

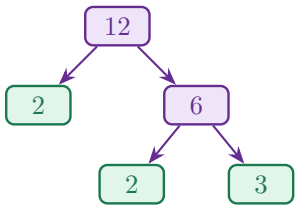
One of the important facts we use all the time is that every number can not only be factorised into various pairs of factors — it can be factorised **uniquely** into the prime numbers that form it.

Example — factorising 12

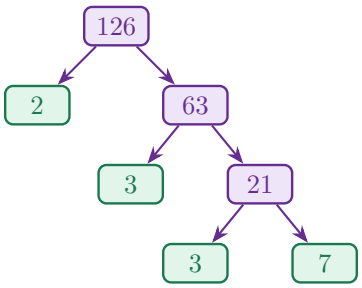
We saw that $12 = 2 \times 6$, and also $12 = 4 \times 3$, and $12 = 1 \times 12$. But fundamentally 12 has three prime factors: 2, 2 again, and 3.

$$12 \;=\; 2 \times 2 \times 3 \;=\; 2^2 \cdot 3$$

Depending on how we combine these primes we recover 4×3 or 2×6, but $2 \times 2 \times 3$ is the one and only way of writing 12 as a product of primes.



$12 = 2 \times 2 \times 3 = 2^2 \cdot 3$



$126 = 2 \cdot 3^2 \cdot 7$

Example — factorising 126

$$126 \;=\; 2 \times 63 \;=\; 2 \times 3 \times 21 \;=\; 2 \times 3 \times 3 \times 7$$

So the prime factors are precisely 2, 3 twice, and 7, and using our exponent notation we condense this to

$$126 \;=\; 2 \cdot 3^2 \cdot 7.$$

Prime factorisation

Every integer can be decomposed into a product of primes in a **unique** way:

$$n \;=\; p_1^{a_1} \cdot p_2^{a_2} \cdots p_r^{a_r}$$

where $p_1, p_2, \ldots, p_r$ are primes. This is called the **prime factorisation** of n. We use it implicitly all the time, and we shall see later how it is put to work.

Lecture 2 — Rational Numbers

Fractions, $\mathbb{Q}$, equivalent forms, reduced form and the gcd, discrete versus dense

What this lecture covers

In the first lecture on numbers we looked at the natural numbers and the integers. Now let us see what happens when we try to *divide*. This leads us to the rational numbers: their definition as ratios of integers, the fact that the same rational number can be written in infinitely many ways, how that fact is used for addition and comparison, the reduced form obtained through the greatest common divisor, and finally the striking difference between the *discrete* integers and the *dense* rationals.

1 From Division to Fractions

We saw that $19 \div 5$ cannot be represented as an integer, because we cannot find a number k such that

$$5 \times k \;=\; 19.$$

The way we deal with this is to represent the quantity as a **fraction**:

$$\frac{19}{5} \;=\; 3\,\frac{4}{5}$$

This number $3\frac{4}{5}$ is an example of a **rational number** — what in school we simply called a fraction.

Rational numbers

A **rational number** is anything that can be written as

$$\frac{p}{q}, \qquad \text{where } p \text{ and } q \text{ are both integers.}$$

The number on the top, p, is called the **numerator**; the number on the bottom, q, is called the **denominator**.
Just as we wrote $\mathbb{N}$ for the natural numbers and $\mathbb{Z}$ for the integers, we write the somewhat unusual symbol $\mathbb{Q}$, with the doubled lines, for the set of rational numbers:

$$\mathbb{Q} \;=\; \left\{ \frac{p}{q} \;:\; p, q \in \mathbb{Z} \right\}$$

Where the name comes from

The word *rational* comes from **ratio**: a rational number is just the ratio of two integers, p divided by q.

2 The Representation Is Not Unique

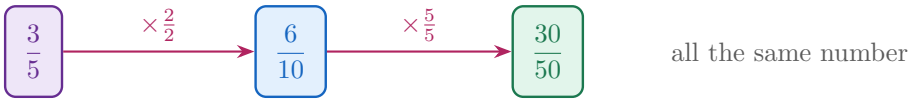
One important feature of the rational numbers is that **the same number can be written in many different ways**. This is not true of the integers: if you write 7, there is only one way to write 7, once the notation for writing numbers is fixed. (We are of course not talking about changing base, from binary to decimal or the like.)

With rationals this fails, because there are many ways of writing p/q that all stand for the same number.

Example — the many faces of 3/5

$$\frac{3}{5} = \frac{3\times 2}{5\times 2} = \frac{6}{10} = \frac{3\times 10}{5\times 10} = \frac{30}{50}$$

Multiplying the top and the bottom by the *same* quantity gives a new fraction that represents exactly the same number.



This is sometimes a nuisance, but it is also sometimes extremely useful, as we now see.

3 Adding and Comparing Fractions

There is no reasonable way to compare two numbers such as $\frac{3}{5}$ and $\frac{3}{4}$, or $\frac{2}{5}$ and $\frac{3}{4}$, directly. If two fractions have *different* denominators, there is no way to compare them as they stand. The only way is to convert them into **equivalent fractions with the same denominator**.

The usual recipe is to find a number into which both denominators divide — that is, a **common multiple** of the two denominators. The smallest such number is called the **least common multiple**, but in fact any common multiple will do.

Example — adding $\frac{3}{5}$ and $\frac{3}{4}$

We cannot add these directly. But 20 is a multiple of both 5 and 4, so

$$\frac{3}{5} \equiv \frac{12}{20}, \qquad \frac{3}{4} \equiv \frac{15}{20}$$

and now that the denominators agree we may simply add the numerators:

$$\frac{3}{5}+\frac{3}{4} = \frac{12}{20}+\frac{15}{20} = \frac{12+15}{20} = \frac{27}{20}$$

Example — comparing $\frac{3}{5}$ and $\frac{3}{4}$

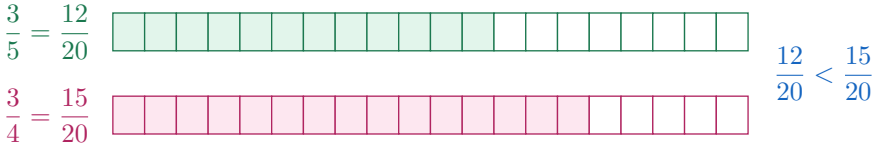
Again we make the denominators the same:

$$\frac{12}{20} < \frac{15}{20} \qquad \Longrightarrow \qquad \frac{3}{5} < \frac{3}{4}$$

The reason is plain: if you divide something into 20 parts and take 12 of them, that is less than taking 15 of them.

The common denominator need not be the smallest one. We could equally well take 100, since 5 and 4 both go into 100:

$$\frac{3}{5} = \frac{60}{100}, \qquad \frac{3}{4} = \frac{75}{100}, \qquad 60 < 75 \implies \frac{60}{100} < \frac{75}{100} \implies \frac{3}{5} < \frac{3}{4}$$



Why non-uniqueness is a blessing

Both addition and comparison require the denominators to be the same. It is therefore extremely useful that the same rational number can be written in many different ways — we simply pick the representation that suits the calculation in hand.

4 The Reduced Form and the Greatest Common Divisor

Since the representation is not unique, how do we choose the *best* way of writing a rational number? If we are not in the middle of an arithmetic operation or a comparison, we prefer the **reduced form**.

Reduced form

A rational number $\frac{p}{q}$ is in **reduced form** if there is no common factor between the numerator and the denominator; that is, there is no factor $f > 1$ with

$$f \mid p \quad \text{and} \quad f \mid q.$$

Equivalently, the reduced form is the one for which

$$\gcd(p,\, q) \;=\; 1.$$

Example — reducing $\frac{18}{60}$

$$\frac{18}{60} \;=\; \frac{3}{10}$$

Notice that $3 = 3 \times 1$ and $10 = 5 \times 2 \times 1$, so there is no common factor between the top and the bottom: the fraction is in reduced form.

This is the **greatest common divisor** problem: we want the largest number that divides both the numerator and the denominator, and then we divide both by it to reach the reduced form. Here

$$\gcd(18,\, 60) \;=\; 6, \qquad \frac{18 \div 6}{60 \div 6} \;=\; \frac{3}{10}.$$

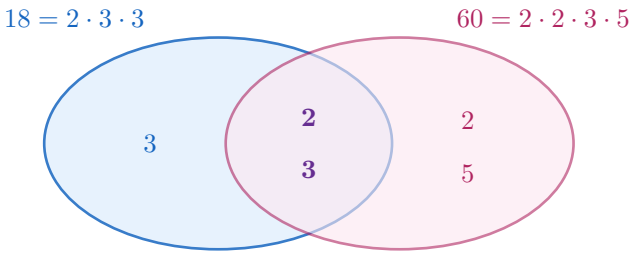
Finding the gcd from the prime factorisation

We can obtain the gcd using the prime factorisation from the previous lecture:

$$18 \;=\; 2 \times 3 \times 3 \qquad (\,2 \times 3 = 6,\; 6 \times 3 = 18\,)$$
$$60 \;=\; 2 \times 2 \times 3 \times 5 \qquad (\,4 \times 3 = 12,\; 12 \times 5 = 60\,)$$

Now look at what is common. There is one 2 in 18 and one 2 in 60, so that 2 is part of a shared factor; there is a 3 in 18 and a 3 in 60, so that too is shared. The second 2 of 60 is not present in 18. Therefore

$$\gcd(18, 60) \;=\; 2 \times 3 \;=\; 6.$$



common primes: $2 \times 3 = 6 = \gcd(18, 60)$

There is no bigger common factor: to assemble one we would have to pull out one more prime from each side, but no prime is left that is present on both sides. The remaining 3 belongs to 18; the remaining 2 and 5 belong to 60; and there is no matching partner on the other side.

The rule for the gcd

Take the prime factorisation of both numbers and collect together all the primes that occur in both, the same number of times. Their product is the gcd.

A remark on efficiency

This is not the best way to find the gcd — there are far more efficient methods — but it tells us intuitively what the gcd actually *is*.

5 Discrete versus Dense

Here is another interesting property of the rational numbers.

Integers are discrete

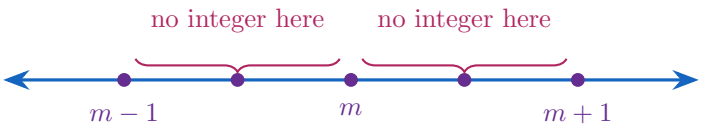
For each integer we know intuitively what the *next* and the *previous* integer is. If I tell you 22 and ask for the next integer, you will say 23; the previous one is 21.

for every integer m : next $= m + 1$, previous $= m - 1$

It does not matter whether the integer is positive or negative:

$17 \longrightarrow$ next 18, previous 16 $\qquad -1 \longrightarrow$ next 0, previous -2

The defining property of *next* is that there is **nothing in between**: there is no integer strictly between m and $m+1$, and none between m and $m-1$. It is the immediate neighbour on the number line, not merely some bigger or smaller integer.



the integers are discrete: there are gaps

Rationals are dense

What about the rationals? Is it possible to speak of the next and the previous rational number? It turns out that this is **not possible**, for a very simple reason: between any two rationals we can always find another one, by taking their **average**.

Recall that the average of any two numbers must lie between them, since it is their sum divided by 2; the average can be neither smaller than both nor bigger than both. If the two numbers differ, the average lies *strictly* between them; if they are the same, the average is that same number.

Example — averages of marks

If somebody scores 37 marks and 37 marks, their average is 37. But if they score 37 marks and 52 marks, then without calculating anything you already know that the average is bigger than 37 and smaller than 52.

Now take two fractions $\frac{m}{n}$ and $\frac{p}{q}$ with

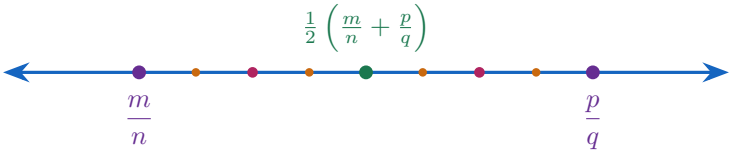
$$\frac{m}{n} < \frac{p}{q}$$

(to establish this we would normally have to bring them to a common denominator, as above). Suppose somebody claims that these two are *adjacent* — that p/q is the next rational after m/n. We answer: take the average of the two.

$$\frac{m}{n} < \underbrace{\frac{1}{2}\left(\frac{m}{n}+\frac{p}{q}\right)}_{\text{a new rational } \frac{a}{b}} < \frac{p}{q}$$

$$\frac{1}{2}\left(\frac{m}{n}+\frac{p}{q}\right) = \frac{mq+np}{2nq} = \frac{a}{b}$$

Working out $\left(\frac{m}{n}+\frac{p}{q}\right) \div 2$ and simplifying gives a new number that is again of the form a/b with a and b integers. So it is itself a rational number, and, as argued, it lies strictly between the two. Hence between any two rational numbers we can always find another one.



and we can keep going for ever — no gaps anywhere

Discrete and dense

- A set is **discrete** if it has the *next* property: every element has an immediate neighbour with nothing in between. The natural numbers $\mathbb{N}$ and the integers $\mathbb{Z}$ are discrete.
- A set is **dense** if it has no *next* property: between any two elements there is always another element. The rational numbers $\mathbb{Q}$ are dense.

Dense here has its usual sense — closely packed together. You cannot find any gaps in the rationals, whereas on the number line there is a genuine gap between m and $m+1$ with no integer in it.

Lecture 3 — Real Numbers

Square roots, $\sqrt{2}$ and the irrationals, $\mathbb{R}$, π and e, and a glimpse of the complex numbers

What this lecture covers

We started with the natural numbers and the integers, and then moved on to the rational numbers p/q. We decided that the rationals are *dense*: between any two of them lies another. It is therefore natural to ask whether the whole number line is made up of rational numbers. In this lecture we find that it is not — there are genuine gaps — and filling them gives us the irrational numbers and hence the real numbers $\mathbb{R}$. We close by asking what happens to the square root of a negative number.

1 Does $\mathbb{Q}$ Fill the Number Line?

The rationals are dense, so on the number line, between any two positions we can find a rational, and between those another, and so on. This makes the following question a sensible one to ask:

Is the entire number line composed only of rational numbers?

Some of those rationals are of course integers. An integer *is* a rational number, since we may write

$$7 = \frac{7}{1},$$

which is of the form p/q. Indeed, any rational number whose reduced form has denominator 1 is an integer, so an integer is a special case of a rational number.

The answer

It turns out that the rationals do **not** fill up the number line. There are points on the line — perfectly measurable lengths — that are not rational numbers at all.

2 Squares and Square Roots

Recall that the square of a number is the number multiplied by itself:

$$m^2 = m \times m.$$

Turning this operation around gives the **square root**.

Square root

The square root of m is that number r for which

$$r \times r = m, \qquad \text{written} \qquad r = \sqrt{m}.$$

In words: which number must we square in order to get m?

If we take the so-called **perfect squares** $1, 4, 9, 16, 25, \ldots$, their square roots are integers:

Examples — square roots of perfect squares

$$1^2 = 1 \implies \sqrt{1} = 1 \qquad 2^2 = 4 \implies \sqrt{4} = 2$$
$$5^2 = 25 \implies \sqrt{25} = 5 \qquad 16^2 = 256 \implies \sqrt{256} = 16$$

So some integers are clearly squares of other integers, and taking the square root returns an integer. But what happens when a number is *not* a perfect square?

Example — $\sqrt{10}$

Take 10, which is not a perfect square. Its square root is not an integer: it lies somewhere between 3 and 4, because

$$3^2 = 9 < 10 < 16 = 4^2 \qquad \implies \qquad 3 < \sqrt{10} < 4.$$

The question is: is it a *rational* number or not?

3 The Square Root of 2

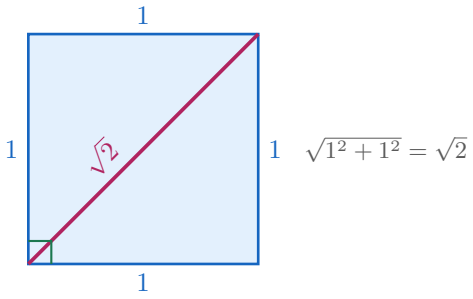
Since $1 = 1 \times 1$ is itself a perfect square, the smallest integer that is not a perfect square is 2. And it is one of the very oldest results in mathematics that

$$\sqrt{2} \neq \frac{p}{q} \qquad \text{for any integers } p, q.$$

This was certainly known to the ancient Greeks, and in particular to Pythagoras.

Yet $\sqrt{2}$ is not an unreal quantity in any sense — we can physically draw a line of that length. Take a square whose sides are of length 1. By Pythagoras' theorem the hypotenuse of the triangle formed by two sides is

$$\sqrt{1^2 + 1^2} = \sqrt{1+1} = \sqrt{2}.$$



So here is a number which is a very measurable quantity — we can draw it as a length — and yet it does not fit into the number line of rational numbers, even though the rationals seemed to cover everything because they are dense.

On the proof

The reasons why $\sqrt{2}$ cannot be written as p/q are not described here; the argument is taken up separately for those who are interested.

4 Irrational Numbers and the Real Numbers

Irrational and real numbers

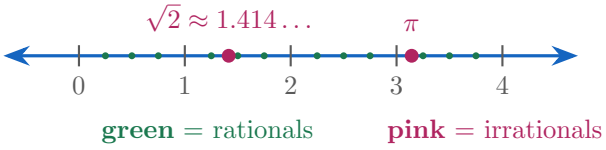
A number such as $\sqrt{2}$, which exists on the number line but is **not** a rational number, is called an **irrational number**.
All the rational numbers together with all the irrational numbers are called the **real numbers**, denoted by the doubled letter $\mathbb{R}$:

$$\mathbb{R} \;=\; \mathbb{Q} \,\cup\, \{\text{irrational numbers}\}$$

So we now have four families of numbers, each with its own symbol:

$$\mathbb{N}\ (\text{natural}), \qquad \mathbb{Z}\ (\text{integers}), \qquad \mathbb{Q}\ (\text{rational}), \qquad \mathbb{R}\ (\text{real}).$$

The real numbers extend the rationals by these irrational numbers, which are very much on the number line but cannot be written in the form p/q.



The real numbers are dense too

It is not difficult to argue that, just like the rationals, the real numbers are dense — and for exactly the same reason. If r and r' are real numbers with $r < r'$, take their average:

$$r \;<\; \frac{r+r'}{2} \;<\; r'$$

This number is bigger than r and smaller than r', so it lies between them. Hence between any two real numbers there is another real number, and $\mathbb{R}$ is dense.

5 Two Famous Irrational Numbers

Some irrational numbers are used a great deal in mathematics, and you have probably come across them already.

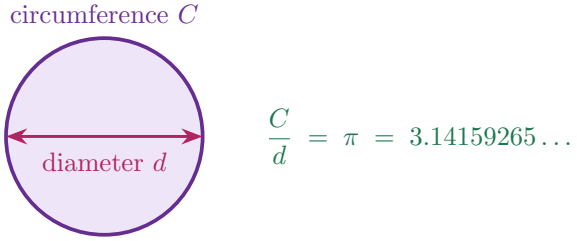
The number π

The number π arises whenever we talk about circles: it is the ratio of the circumference to the diameter,

$$\pi \;=\; \frac{\text{circumference}}{\text{diameter}},$$

and this is an **invariant** — for *any* circle whatsoever, the circumference divided by the diameter is π. It cannot be written in the form p/q, and its decimal expansion runs on for ever:

$$\pi \;=\; 3.14159265\ldots$$



The number e

Another very popular irrational number is e, which is used for natural logarithms:

$$e \;=\; 2.7182818\ldots$$

Square roots are almost always irrational

As we have seen, $\sqrt{2}$ is an irrational number — and so is $\sqrt{3}$, and $\sqrt{5}$, and $\sqrt{6}$, and so on:

$$\sqrt{n} \text{ is irrational for every integer } n \text{ that is not a perfect square.}$$

Except for the perfect squares, none of the square roots of integers are rational. Many of these numbers are not especially useful to us, but π and e certainly are.

6 Do We Stop Here? The Square Root of a Negative Number

We have found more numbers on the line than just the rationals, namely the reals. Do we stop here? Let us look again at the square root operation, which is what led us to the irrationals in the first place. What happens if we take the square root of a **negative** number, such as -1?

Recall the sign rule for multiplication from the first lecture: if the two signs are the same — two negatives or two positives — the answer is positive; only if the two signs differ do we get a negative answer.

$$(+)\times(+) \;=\; (+) \qquad (-)\times(-) \;=\; (+) \qquad (+)\times(-) \;=\; (-)$$

So to multiply two numbers and obtain -1, one of them must be negative and the other must not be. But by definition a square root is a number multiplied *by itself* — the same number, and so necessarily the same sign. Any square root multiplied by itself must therefore give a **positive** number:

$$r \times r \;=\; -1 \qquad \text{is impossible for any real } r.$$

Hence a negative number has no square root among the reals.

Complex numbers

To find square roots of negative numbers we must create yet another class of numbers, called the **complex numbers**. Complex numbers extend the real numbers, just as the reals extend the rationals and the rationals extend the integers. The good news is that we do not need complex numbers for this course.

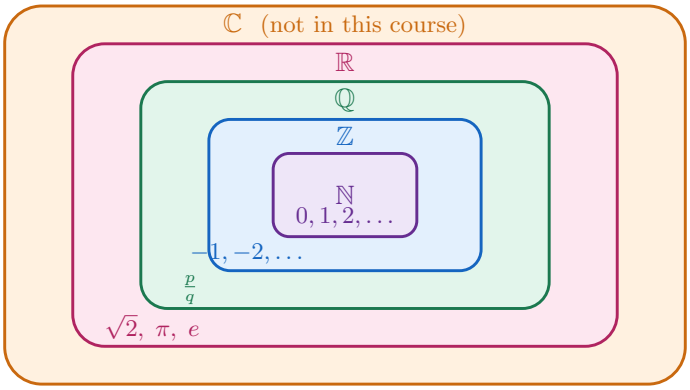
7 The Progression of Number Systems

Putting everything together, we have a chain of number systems, each containing the one before it:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

- Every **natural number** is also an **integer**, because the integers extend the naturals with the negative quantities.
- Every **integer** is also a **rational number**, because we can think of every integer as a ratio p/q with denominator 1.
- Every **rational number** is also a **real number**, because the reals consist of all the rationals plus all the irrationals.

- Beyond the reals lie the **complex numbers**, which we shall not discuss.



Lecture 4 — Sets

Collections, cardinality, elements and subsets, Venn diagrams, the empty set and the power set

What this lecture covers

We have seen the natural numbers, the integers, the rationals and the reals, and have loosely called them *sets* of numbers. Let us now understand a little more clearly what a set is: how we describe one, how we count its members, what it means for one set to sit inside another, and finally how to collect all the subsets of a set into a new set of its own.

1 What Is a Set?

At its most basic level, a **set** is a collection of items.

Examples of finite sets

$$\text{Days of the week} = \{\text{Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday}\}$$

$$\text{Factors of } 24 = \{1,\ 2,\ 3,\ 4,\ 6,\ 8,\ 12,\ 24\} \quad \text{— there are 8 of them}$$

$$\text{Primes below } 15 = \{2,\ 3,\ 5,\ 7,\ 11,\ 13\}$$

Recall that 1 is not a prime; $2, 3, 5, 7$ are the single-digit primes, and then come 11 and 13.

Sets can of course be infinite, and the infinite sets we deal with most commonly are the different families of numbers:

Set	Description
$\mathbb{N}$	the natural numbers $0, 1, 2, 3, 4, \ldots$
$\mathbb{Z}$	the integers: the naturals together with $-1, -2, -3, \ldots$
$\mathbb{Q}$	the rationals: the fractions p/q with p, q both integers
$\mathbb{R}$	the reals: all the rationals, plus numbers such as $\sqrt{2}$, π and e that cannot be written as fractions

There is no uniformity of type

In all the examples above it looks as though a set must have some uniformity — either it consists of numbers, or of days of the week, or of something similar. Mathematically there is **no such constraint**. A set may have any kind of members, even a mixed membership.

Example — the objects in a painting

Consider enumerating the objects that appear in a painting — for instance Pablo Picasso's *Three Musicians*, where the abstract style makes the enumeration rather hard. Roughly, we can make out three people, some musical instruments, and, if we look very carefully, even a dog.
Notice that there is no commonality at all among the members: there are people, musical instruments, chairs, tables and animals. A set may perfectly well have members of mixed type.

Order and repetition do not matter

An important difference between a set and a sequence or a list is that the **order** in which we identify the members does not matter. Nor does it matter if we accidentally write the same member twice.

Example — a set of cricketers

$$\{\text{Kohli, Dhoni, Pujara}\} = \{\text{Pujara, Kohli, Dhoni}\} = \{\text{Kohli, Dhoni, Pujara, Kohli}\}$$

All three describe the same set — though of course, if you are a cricket fan, you might well like Kohli to bat twice for you.

2 Cardinality

Cardinality

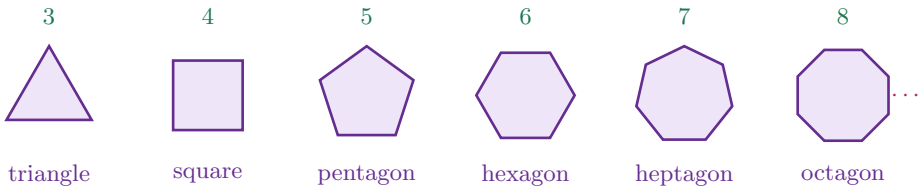
The **cardinality** of a set is the number of items in it. For a finite set we can simply count the members.

For example, the factors of 24 listed above give

$$\left|\{1, 2, 3, 4, 6, 8, 12, 24\}\right| = 8.$$

When finiteness is not obvious

Sometimes it is not obvious that a set is finite at all. Recall from geometry that a **regular polygon** is one in which all the sides are equal and all the angles are equal. The smallest regular polygon is the equilateral triangle, with 3 equal sides and 3 internal angles of 60° each. Then come the square, the regular pentagon, the hexagon, the heptagon, the octagon, and so on.



For *any* number of sides you can draw a regular polygon with that many sides and equal interior angles. There is no limit, so the set of regular polygons is **infinite**.

But if we move to three dimensions, the corresponding notion is the **Platonic solid**: a solid each of whose faces is a regular polygon, with all these polygons meeting at the same angle in three dimensions. Although there are infinitely many regular polygons in two dimensions, it turns out that there are only **five** Platonic solids in three dimensions.

The five Platonic solids

Solid	Faces	Shape of each face
Tetrahedron	4	triangles
Cube	6	squares
Octahedron	8	triangles
Dodecahedron	12	pentagons
Icosahedron	20	triangles

There are no other regular solids — a surprising fact. This is an example of a set that turns out to be finite even though there is no obvious reason for it to be so.

Cardinality of infinite sets

Cardinality is easy to determine for a finite set, but what about an infinite one? Recall that we moved from the integers to the rationals because dividing two integers need not give an integer; the integers are *discrete* (there is a next and a previous, so there are gaps) whereas the rationals are *dense*. Intuitively it seems that we are adding things to $\mathbb{Z}$ to get $\mathbb{Q}$. But can we make this formal?

Are there more rational numbers than there are integers? And is $\mathbb{R}$ really bigger than $\mathbb{Q}$?

Clearly some real numbers, such as $\sqrt{2}$, are not rational, so we do have a bigger set in that sense. There is indeed a way to measure the cardinality of infinite sets, but it is not as straightforward as for finite sets; this is taken up in a separate discussion.

3 Describing a Set

For a finite set we can simply list its members explicitly. The normal notation is to enclose the list in **curly braces** and separate the items by commas:

$$\{\text{Kohli, Dhoni, Pujara}\}, \qquad \{1, 2, 3, 4, 6, 8, 12, 24\}.$$

In many books — and in these lectures — we also see notation such as

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

indicating an infinite collection of elements following some pattern.

The dots are informal

This looks like a way of listing an infinite set, but it must be understood that it is **only an informal notation**, not a formal one. You cannot write "$\dots$" and claim to have listed an infinite set; some other way of describing it is needed.

Not every collection is a set

It seems reasonable that if a set is a collection of items, then we may collect anything at all and call it a set. This turns out not to be quite true, and the problem shows up particularly with infinite sets. Bertrand Russell showed that there is a genuine problem if we collect *all sets* together and call that a set.

Russell's paradox

A "set of all sets" leads to a contradiction, known as **Russell's paradox**, which is discussed separately. So although the notion of a set is intuitive, and it seems natural that any collection of objects should be a set, we must be a little careful in mathematics about what is and is not a set.

For everything we shall see in this course the situation is straightforward: whenever we have a collection of numbers, or of objects with a mathematical description, we may safely assume it is a set.

4 Elements and Membership

The most formal word for a member of a set is an **element**: a set consists of elements. We write membership using the symbol $\in$, read as "is an element of".

Membership notation

$$x \in X \qquad \text{means} \qquad x \text{ is a member of the set } X$$

The same symbol with a stroke across it, $\notin$, means "is not an element of".

Examples of membership

$$0 \in \mathbb{N} \qquad 5 \in \mathbb{Z} \qquad \sqrt{2} \notin \mathbb{Q}$$

The last one records our earlier claim: $\sqrt{2}$ is not a rational number.

5 Subsets

Moving on from elements, we can compare sets by asking whether one is included in another.

Subset

X is a **subset** of Y if every element of X is also an element of Y:

$$X \subseteq Y \qquad \Longleftrightarrow \qquad \text{for every } x,\ x \in X \implies x \in Y.$$

Examples of subsets

Taking two of the three players listed earlier:

$$\{\text{Kohli, Pujara}\} \subseteq \{\text{Kohli, Dhoni, Pujara}\}$$

Collecting only the prime numbers out of the natural numbers (recall that a prime p has exactly two factors, 1 and p), and noting that many numbers are not prime:

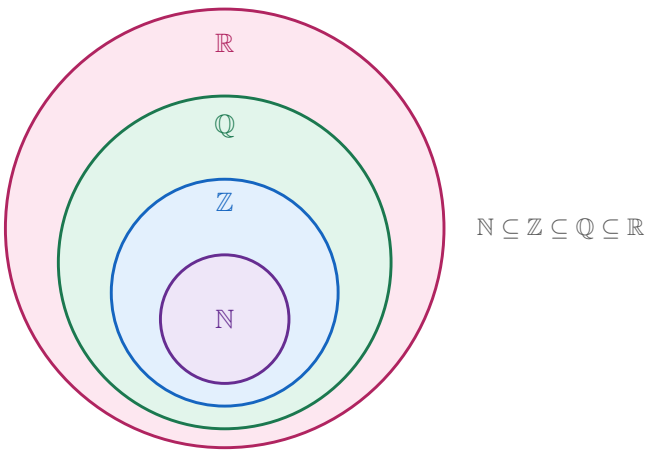
$$\{\text{primes}\} \subseteq \mathbb{N}$$

Since the integers extend the naturals by the negative numbers, and $\mathbb{Z}$ was extended to $\mathbb{Q}$, which was extended to $\mathbb{R}$:

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

Venn diagrams

We can draw this by representing each set as a region: a large circle for the reals, and smaller circles inside it. If one circle lies inside another, the inner set is a subset of the outer one.



This is a very abstract diagram: we are not listing the elements of the sets, only indicating their extent — that $\mathbb{R}$ reaches beyond $\mathbb{Q}$, and that everything in $\mathbb{Q}$ sits inside $\mathbb{R}$. Such pictures are called **Venn diagrams**, and they are a very useful way of visualising sets and the relationships between them.

Every set is a subset of itself

Notice that, by the definition, every set is a subset of itself: since every element of X is also an element of X, we trivially have

$$X \subseteq X.$$

This gives us an important notion that looks obvious but is not always so.

Equality of sets

Two sets are equal if and only if they consist of exactly the same elements. One way to check this is:

$$X = Y \qquad \Longleftrightarrow \qquad X \subseteq Y \text{ and } Y \subseteq X$$

This is used whenever we have two *different descriptions* of the same collection of objects and wish to check whether they agree: we argue that everything satisfying the first description also satisfies the second, and vice versa. Although it looks obvious for finite sets, for infinite sets we sometimes have to argue in this indirect way, which is why the statement matters.

Proper subsets

Sometimes we want to distinguish the case where X is genuinely smaller than Y from the case where it might be equal to it. The notation $\subseteq$ allows *both*: when we write $X \subseteq Y$ we are saying X is a subset, possibly even equal to Y.

For **proper subsets** — where X does not include all of Y — we use a different notation: either drop the equality sign and write $\subset$, or strike the equality out explicitly and write $\subsetneq$.

A word of caution

The second symbol, $\subsetneq$, is always unambiguous. The bare $\subset$ is used both ways in the literature, so when reading a book you must be a little careful about whether the author means "subset" or "proper subset".

Examples of proper subsets

$\mathbb{N} \subsetneq \mathbb{Z}$ — the negative numbers are not in $\mathbb{N}$

$\mathbb{Z} \subsetneq \mathbb{Q}$ $\qquad$ $\mathbb{Q} \subsetneq \mathbb{R}$ — the irrationals are not rational

In most interesting cases we will be looking at proper subsets. Sometimes we emphasise this by striking out the equality sign, and sometimes we will not; very often the context makes it clear which is meant.

6 The Empty Set

Just as 0 is a very important number, there is a very important set in set theory which plays the equivalent role: the set with no elements.

The empty set

The set with no elements is called the **empty set** and is written

$$\varnothing \qquad \text{(a zero with a line across it).}$$

It has no elements at all.

What may not be obvious is that the empty set is a subset of *every* set:

$$\varnothing \subseteq X \qquad \text{for every set } X.$$

One might object that the empty set has no elements, so how can the condition hold? The answer is that when we say "for every" and there is nothing in the set, the statement is automatically true.

Example — birds with three legs

If I say "all birds with three legs have pink beaks", this is actually a true statement. There are no birds with three legs, so every bird that does have three legs has a pink beak — vacuously.

Such statements are called **vacuously true**, and they hold for any sentence using the word "all" when the set in question is empty. In particular, every element that could be in the empty set is also an element of any set X you care to build, so the empty set is a subset of every possible set.

7 The Power Set

So far we have treated elements and sets as two different categories of objects. But there is no restriction requiring the elements of a set to be indivisible objects: **a set can perfectly well contain other sets**.

One of the important sets of sets is the **power set**: the set of all subsets of a given set.

Example — the power set of $\{a, b\}$

Take the two-element set $X = \{a, b\}$. What are all its subsets?

- $\varnothing$ — the empty set is always a subset;
- $\{a, b\}$ — the set itself, since $X \subseteq X$ for any X;
- $\{a\}$ — include a, exclude b;
- $\{b\}$ — include b, exclude a.

There are four subsets, and grouping them together into a larger set gives the power set:

$$\mathcal{P}(X) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}$$

$\{\varnothing\}$ is not the same as $\varnothing$

$$\{\varnothing\} \neq \varnothing$$

The first is a set consisting of *one* element, namely the empty set. The second is the empty set itself, a set with *no* elements. Putting a brace around the empty-set symbol creates a set with one element.

Example — the power set of the empty set

What is $\mathcal{P}(\varnothing)$? The empty set has the empty set as a subset, and there is nothing else — the

full set is also the empty set, and duplicating an element changes nothing. So

$$\mathcal{P}(\varnothing) \;=\; \{\,\varnothing\,\}$$

a set consisting of just one element, namely the empty set itself.

How many subsets are there?

With two elements we found four subsets. This generalises:

a set with n elements has exactly 2^n subsets.

Example — the subsets of $\{a, b, c\}$

$$\underbrace{\varnothing}_{1},\qquad \underbrace{\{a\},\ \{b\},\ \{c\}}_{3},\qquad \underbrace{\{a,b\},\ \{a,c\},\ \{b,c\}}_{3},\qquad \underbrace{\{a,b,c\}}_{1}$$

These are the only subsets. Adding them up:

$$1+3+3+1 \;=\; 8 \;=\; 2^3.$$

You can check that for $\{a, b, c, d\}$ you would get $2^4 = 16$.

Why 2^n? The counting argument

Suppose the set has n elements, which we call $x_1, x_2, \ldots, x_n$ without describing what they are. These must all be distinct, since elements cannot be duplicated in a set.

Now, how do we construct a subset? For each element x_i we must either **include** it or **exclude** it. So we have to make n choices, each with two options:

$$\underbrace{2 \times 2 \times \cdots \times 2}_{n \text{ times}} \;=\; 2^n$$

We have two ways of deciding what to do with x_1 — keep it or leave it; two for x_2; so 2×2 choices for x_1 and x_2 together, $2 \times 2 \times 2$ for x_1, x_2, x_3 together, and so on. Each different choice generates a different subset, because whenever we make a different decision we either leave out x_i or put it in. Hence there are exactly 2^n subsets.

Another way: subsets as binary numbers

Here is a second way of looking at subsets that reaches the same result. Take again the n-element set $x_1, \ldots, x_n$ and consider n-digit binary numbers. (The word *digit* comes from the decimal system, so for binary we say **bit**, short for binary digit.)

In the binary number system we have only 0s and 1s, and the place values are powers of two — the units place, then the number of twos, of fours, of eights, and so on, exactly as the decimal system works in base 10:

$$\text{base } 10: \quad 10^0,\ 10^1,\ 10^2, \ldots \qquad\qquad \text{base } 2: \quad 2^0,\ 2^1,\ 2^2, \ldots$$

For 3-bit binary numbers there are 8 of them:

$$000,\ 001,\ 010,\ 011,\ 100,\ 101,\ 110,\ 111$$

and the reason there are 2^n of them is again that for each bit we may choose 0 or 1 — two choices for the first bit, two for the second, and so on. So an n-bit binary number can represent 2^n different values, from 0 to $2^n - 1$.

Now we read these n-bit numbers as representing *subsets*. Let the i-th bit record the choice we made about x_i:

bit $= 0 \Rightarrow$ **leave out** x_i

bit $= 1 \Rightarrow$ **keep** x_i

Example — the sequence 0101 over $\{a, b, c, d\}$

Take the four-element set $\{a, b, c, d\}$ and the bit sequence 0 1 0 1:

0	1	0	1
a	b	c	d
leave out	keep	leave out	keep

So the sequence says: leave out a, keep b, leave out c, keep d. Hence

$$0101 \quad \longleftrightarrow \quad \{\, b,\ d \,\}$$

The two extreme sequences

$0000 \quad \longleftrightarrow \quad \varnothing$ — every x_i is omitted, so the subset has no elements

$1111 \quad \longleftrightarrow \quad \{a, b, c, d\}$ — every x_i is included, so we get the set itself

These are the two extreme subsets; all the others lie in between.

Every n-bit number therefore represents one sequence of choices, and there are precisely 2^n such numbers. So any finite set with n elements has exactly 2^n subsets — no more, no less.

Lecture 5 — Set Comprehension, Intervals and Set Operations

Describing subsets of infinite sets; open and closed intervals; union, intersection, difference and complement

What this lecture covers

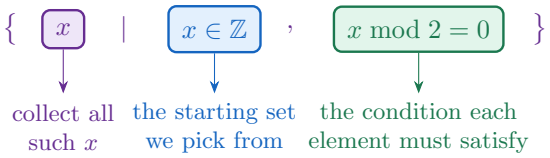
We cannot list out the elements of an infinite set, so how do we describe a subset of the numbers precisely? The answer is **set comprehension**. We use it to define even integers, perfect squares, rationals in reduced form and intervals of numbers, and then turn to the standard operations that combine sets: union, intersection, set difference and complement.

1 Set Comprehension

Suppose we want to talk about the set of **even integers**. The even integers are those integers which, when divided by 2, leave a remainder of 0. Recall that the remainder is written with `mod`, so $x \bmod 2$ is the remainder when x is divided by 2; if $x \bmod 2 = 0$ there is no remainder and x is even.

$$\{\, x \mid x \in \mathbb{Z},\ x \bmod 2 = 0 \,\}$$

Let us separate out the different parts of this notation and see what is going on.



Read aloud, the expression says: collect together all the x such that x is in $\mathbb{Z}$ and $x \bmod 2 = 0$ — in other words, all the even integers.

You must start from an existing set

Set comprehension can only be used when we already have a **starting set**. We begin with a set — $\mathbb{Z}$, $\mathbb{N}$, $\mathbb{Q}$ or $\mathbb{R}$, whose elements we already know — and construct a subset of it. What the comprehension supplies is not the elements but a *description* of how to choose the ones with a given property. This is formally how one defines a subset of an infinite set, since its elements cannot be listed out.

2 More Examples of Set Comprehension

Perfect squares

An integer is a perfect square if its square root is also an integer. For instance 25 is a perfect square because $\sqrt{25} = 5$, but 26 is not, since no integer multiplied by itself gives 26.

Note first that a square must be positive: a negative number cannot be a square, because when we multiply a number by itself the two factors have the same sign, and

$$(-) \times (-) = (+), \qquad (+) \times (+) = (+).$$

So instead of the integers it suffices to look at the natural numbers:

$$\{\, m \mid m \in \mathbb{N},\ \sqrt{m} \in \mathbb{N} \,\} = \{\, 1,\ 4,\ 9,\ 16,\ 25,\ \ldots \,\}$$

The comprehension on the left is a succinct way of writing the informal infinite list on the right: we pull the numbers out of $\mathbb{N}$ one by one, check whether each is a perfect square, and enumerate those that are.

Rationals in reduced form

We saw that the same rational number can be written in many ways, since multiplying numerator and denominator by the same quantity does not change the number, and we used this to make denominators equal for comparison and for addition. Which rationals, then, are actually in reduced form?

Every rational is a pair — a numerator and a denominator, both integers — so it looks like p/q. But we do not want just any p/q: we want those with no common divisor other than 1. Recalling that the gcd is the largest number dividing both p and q:

$$\left\{ \frac{p}{q} \;\middle|\; p, q \in \mathbb{Z},\ \gcd(p, q) = 1 \right\}$$

Example — reduced or not

$\frac{3}{5}$ is in reduced form $\frac{6}{10}$ is not, since $\frac{6}{10} = \frac{3}{5}$

In the second case we can cancel the common factor 2 from top and bottom.

3 Intervals

One thing we often want with numbers is an **interval**: the numbers lying between one value and another.

Intervals of integers

Example — the integers from -6 to 6

$$\{\, z \mid z \in \mathbb{Z},\ -6 \leq z \leq 6 \,\}$$

This says: take all z in the set of integers such that z is greater than or equal to -6 and less than or equal to 6. We could equally well split the condition in two and write $z \geq -6$ and $z \leq 6$ separately.

Syntax is not the point

The way we write the condition may vary, and several forms may be equivalent. We shall not be pedantic about it — in the earlier example we could just as well have written "x is even" instead of $x \bmod 2 = 0$. What matters is the **format**: take an underlying set, run through its elements, and keep those that satisfy the condition.

Intervals of real numbers

Intervals become more interesting with the real numbers, and one we shall want very often is the interval between 0 and 1 — we will frequently talk about probabilities, for instance, and probabilities range between 0 and 1.

Closed interval

Taking all the real numbers between 0 and 1 *including* both endpoints gives the **closed interval**:

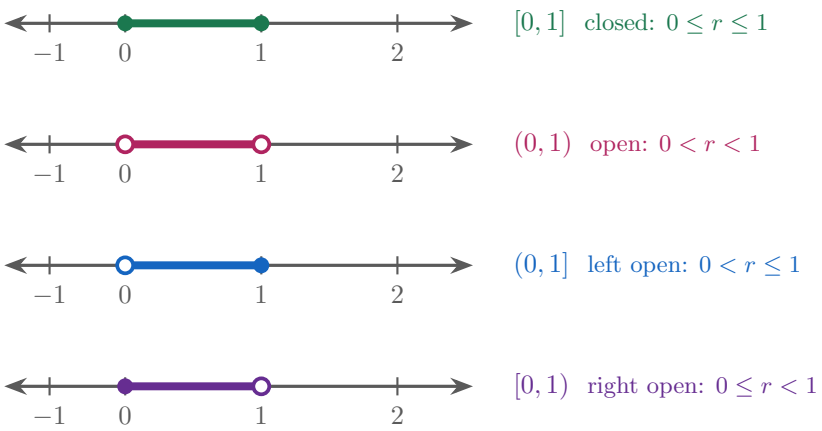
$$[0,1] = \{r \mid r \in \mathbb{R},\ 0 \leq r \leq 1\}$$

Open interval

Excluding both endpoints gives the **open interval**:

$$(0,1) = \{r \mid r \in \mathbb{R},\ 0 < r < 1\}$$

Pictorially we mark an excluded endpoint by an unfilled circle and an included endpoint by a filled one.



There is nothing to stop us from including one endpoint and excluding the other.

Half-open intervals

$$(0,1] = \{r \mid r \in \mathbb{R},\ 0 < r \leq 1\} \quad \textbf{left open}$$

$$[0,1) = \{r \mid r \in \mathbb{R},\ 0 \leq r < 1\} \quad \textbf{right open}$$

Notice the bracket convention: a **round bracket** for an open end, a **square bracket** for a closed end.

Open and closed is just a matter of the inequality

Formally an interval is simply a set defined by set comprehension, and whether it is open or closed depends only on whether the inequality carries an equality sign — strictly less than versus less than or equal to, strictly greater than versus greater than or equal to. Intervals will feature in many later discussions, so it is worth being comfortable with them as subsets of $\mathbb{R}$.

4 Union

Let us now look at some simple operations on sets.

Union

The **union** of two sets combines them into a single set, written $X \cup Y$:

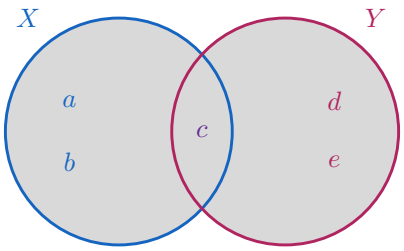
$$X \cup Y = \{x \mid x \in X \text{ or } x \in Y\}$$

Example — union of two three-element sets

$$\{a, b, c\} \cup \{c, d, e\} = \{a, b, c, d, e\}$$

Elements appearing in both sets show up only once in the result, since a set has no duplicates. So the cardinality of the union is in general *less* than the two cardinalities added together: here two three-element sets give a **five**-element set, not a six-element one, because c is common to both.

Returning to Venn diagrams: the picture below shows a general pair of sets X and Y, where X is not a subset of Y and Y is not a subset of X, but there is some overlap. In general some elements belong only to X, some only to Y, and some to both — so two overlapping circles are the general picture of two sets.



the shaded area is $X \cup Y$

Here a, b, c belong to the left circle and c, d, e to the right one, and c is placed in the portion covered by both circles to show that it lies in the common part.

5 Intersection

The corresponding operation that keeps only the elements occurring in *both* sets is the **intersection**, written with the union sign turned upside down.

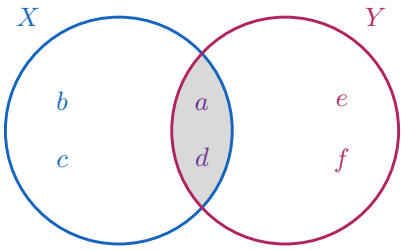
Intersection

$$X \cap Y = \{\, x \mid x \in X \text{ and } x \in Y \,\}$$

Example — intersection

$$\{a, b, c, d\} \cap \{a, d, e, f\} = \{a, d\}$$

Here a is common to both; b and c are not on the right-hand side; d is common to both; and e and f are not on the left-hand side. So only a and d survive.



the shaded area is $X \cap Y$

The shaded region overlapped by both circles is the intersection, and the unshaded portions are those elements that lie in one set but not the other. Note that when we place elements in a Venn diagram the position within a region does not matter, nor does the order.

6 Set Difference

In **set difference** we take two sets and ask what is in the first set that is not in the second.

Set difference

$$X - Y \;=\; X \setminus Y \;=\; \{\, x \mid x \in X \;\text{ and }\; x \notin Y \,\}$$

Both the ordinary minus sign and the backslash are used for this.

Examples with numbers

$\mathbb{R} \setminus \mathbb{Q}$ — the real numbers that are not rational

$\mathbb{Q} \setminus \mathbb{Z}$ — the rational numbers that are not integers

Example — difference is not symmetric

$$\{a, b, c, d\} \setminus \{a, d, e, f\} \;=\; \{b, c\}$$

Here a is in the first set but also in the second, so it is not counted; b and c are in the first set only, so they survive; d appears in both, so it is dropped.
Reversing the order:

$$\{a, d, e, f\} \setminus \{a, b, c, d\} \;=\; \{e, f\}$$

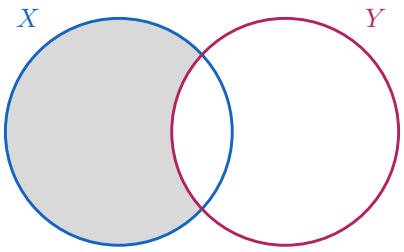
The same common elements a and d disappear, but now what survives is e and f.

Order matters — and when it does not

Set difference behaves like subtraction, where $3-5 \neq 5-3$, whereas addition has $3+5 = 5+3$:

$$X \cup Y \;=\; Y \cup X, \qquad X \cap Y \;=\; Y \cap X, \qquad \textbf{but} \qquad X \setminus Y \;\neq\; Y \setminus X$$

For union and intersection it does not matter which side we take from, since in the end we look at elements included in both sides or in either side.



the shaded area is $X \setminus Y$: everything in X with the overlap removed

7 Complement

Finally we often speak of the **complement** — everything else.

Consider the numbers that are *not* prime. Such numbers are called **composite**: a composite number is one which has factors other than 1 and itself, so it has more than two factors. Clearly a number is either prime or it is not, so the composite numbers are disjoint from the primes and comprise all the numbers that are not prime.

Complement with respect to what?

Complement is not straightforward in set theory, because we must ask: complement with respect to *what*? If we take the complement of the primes inside $\mathbb{R}$, it would include π, e, $\sqrt{2}$ and so on — which is certainly not what we mean. When we say "the complement of the primes" we are thinking of the natural numbers that are not prime.
So we must always specify a **universe** with respect to which we complement.

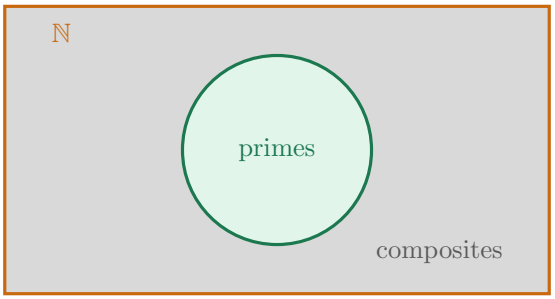
Complement

Given a universe U and a set $X \subseteq U$, the complement of X is

$$\overline{X} \;=\; U \setminus X \;=\; \{\, x \mid x \in U,\; x \notin X \,\}$$

For example, the complement of the primes in the universe of natural numbers gives the composite numbers:

$$\{\text{composites}\} \;=\; \mathbb{N} \setminus \{\text{primes}\}$$



the shaded area is the complement of the primes in the universe $\mathbb{N}$

Whenever the word complement is used, you must make sure you have specified the overall set with respect to which the negation is being taken. If the box above were $\mathbb{R}$ instead of $\mathbb{N}$, we would find $\sqrt{2}$, e and the like sitting in the shaded region, which is not what is intended.

Where sets get used

Sets are the standard way to represent a collection of mathematical objects, so it is worth being comfortable with the terminology ($\in$, $\subseteq$) and the notation (curly braces, set comprehension). Infinite sets — which are the tricky and interesting ones — arise constantly because we think of sets in terms of numbers. But finite collections matter too from a computational and data science point of view: the set of stations at which a train stops in a timetable, or the set of items a store has in its inventory.

Lecture 6 — Set Comprehension in Detail

Generator, filter and transformation; worked examples with squares, cubes and perfect squares

What this lecture covers

Having defined sets and some operations on them, we now look at more examples to become familiar with the notation and terminology. The centrepiece is the anatomy of a set comprehension — the **generator**, the **filter** and the **transformation** — and how the same set can often be described in several different ways depending on which of the three does the work.

1 A Quick Review

A set is a collection of items. If it is **finite** we can simply enumerate the items within curly braces. If it is **infinite** we cannot really write out all the elements, even though informally we put "..." to indicate a continuing sequence — and if that sequence is not very regular, as with the prime numbers, which have no clear pattern, even that is not easy.

The two basic relationships

Membership. A small x typically denotes an element and a capital X the set itself:

$$5 \in \mathbb{Z} \qquad \sqrt{2} \notin \mathbb{Q}$$

Subset. One set is included in another when everything belonging to X belongs to Y:

$$\{\text{primes}\} \subseteq \mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

We draw these with Venn diagrams, using ovals, circles or boxes to represent the extent of a set. Whether one box sits inside another, or merely overlaps it, indicates whether the first set is a subset of the other or whether they simply share some elements.

The power set, recalled

Taking a set, we can enumerate all of its subsets. The **empty set** $\varnothing$ has no elements — we need it for technical reasons — and is a subset of every set; and every set is a subset of itself. For the two-element set $\{a, b\}$:

$$\mathcal{P}(\{a,b\}) \;=\; \{\,\varnothing,\ \{a\},\ \{b\},\ \{a,b\}\,\} \qquad \text{— that is } 2^2 = 4 \text{ subsets.}$$

In general, a finite set with n elements always has 2^n subsets.

2 The Anatomy of a Set Comprehension

Let us now look closely at the set comprehension notation, which is what we use to describe infinite sets that cannot be written down explicitly.

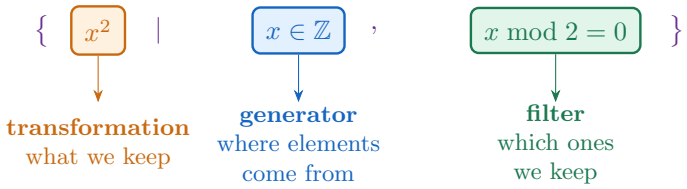
Example — the squares of the even integers

The even integers are $0, \pm 2, \pm 4, \ldots$, and squaring them gives

$$\{\, x^2 \mid x \in \mathbb{Z},\ x \bmod 2 = 0 \,\} \;=\; \{\, 0^2,\ 2^2,\ 4^2,\ 6^2,\ \ldots \,\}$$

Note that $(-2)^2$ is the same as $2^2 = 4$, so the negative even numbers contribute nothing new.

Let us break this up into its parts.



The three parts

- **Generator.** We take elements from an *existing* set — we can only build new sets from old ones. Here $x \in \mathbb{Z}$ says: try every x in $\mathbb{Z}$, so $\mathbb{Z}$ generates the candidates.
- **Filter.** Not all the candidates are interesting, so we keep only those satisfying a given property. Here the property is $x \bmod 2 = 0$, that is, x is even.
- **Transformation.** The surviving elements need not be kept as they are. The expression on the *left* of the vertical bar gives the actual elements of the set. Here we want the squares, not the even numbers themselves, so we write x^2.

So the elements are *generated*, then *filtered* to rule out the ones we do not want, and then the survivors are *transformed*.



Notice that $(-2)^2 = 4$ and $2^2 = 4$, so some elements disappear in the final set — we do not keep duplicates.

Writing the filter

Filters may be written in many different ways, as long as they are unambiguous. We shall not be particular about the language used, so long as there is no question about what is meant.

3 More Examples of the Three Parts

Rationals in reduced form — the identity transformation

Some rational numbers are not in reduced form. For instance, $\frac{4}{10}$ should really be thought of as $\frac{2}{5}$, since

$$\frac{2}{5} \times \frac{2}{2} = \frac{4}{10}.$$

We have multiplied numerator and denominator by 2 to go from $\frac{2}{5}$ to $\frac{4}{10}$, but it is the same rational number. We want the numerator and denominator to have no common divisor, which is the same as saying their greatest common divisor is 1.

$$\left\{ \frac{p}{q} \;\middle|\; \frac{p}{q} \in \mathbb{Q},\ \gcd(p,q) = 1 \right\}$$

Here we generate all the possible rationals p/q, filter out those whose numerator and denominator share a divisor, and then keep the survivors exactly as they are. This is the **identity transformation** — the identity simply takes an input and produces the same thing as output.

Using gcd as a function

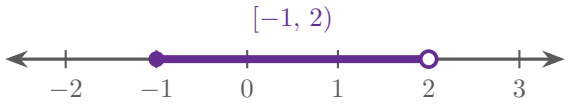
We have used a function, gcd, without formally defining it here; we assume the reader understands what it means. This is exactly what is meant by saying that the filter may be written in any reasonable way, as long as people understand it.

An interval

Suppose we want the real numbers starting from -1, including -1, and going up to but *not* including 2:

$$[-1, 2) \;=\; \{\, r \mid r \in \mathbb{R},\; -1 \le r < 2 \,\}$$

We take every possible real number, then check that it is greater than or equal to -1 — cutting off everything strictly smaller — and that it does not reach 2. Again there is no transformation. The square and round brackets in the notation on the left indicate whether an endpoint is included: -1 is in, $+2$ is out.



4 Why We Want Set Comprehension

Let us extend the first example from squares to **cubes**. A cube is a number multiplied by itself three times: a square is $x \times x$, a cube is $x \times x \times x$.

If we want the cubes of the first five natural numbers, we can write the generator out explicitly:

$$\{\, n^3 \mid n \in \{0, 1, 2, 3, 4\} \,\}$$

Natural numbers start at 0

In our terminology the natural numbers start with 0, so the first five of them are $0, 1, 2, 3, 4$ — even though in some books the naturals begin at 1.

Here we take every n in the generating set and transform it to n^3, with no filtering at all: we are not asking for the first five odd numbers, or for numbers with some other property, just the first five numbers.

Now make it 500

Imagine we change the question to the cubes of the first **500** natural numbers. We could still write the generator out explicitly, but it would be extremely tedious — a list of 500 numbers instead of 5 — and remember that we are not really allowed to write "…" if we are being mathematically precise. Set comprehension solves this neatly:

$$X \;=\; \{\, n \mid n \in \mathbb{N},\; n < 500 \,\} \;=\; \{\, 0, 1, 2, \ldots, 499 \,\}$$
$$\{\, n^3 \mid n \in X \,\}$$

The first line replaces a very long and tedious expression by a compact one capturing exactly the same set; the second uses it as a generator. Since X is restricted to 0 through 499, we get the cubes over exactly that range. So a second use of set comprehension is to make definitions more readable, more understandable and less tedious to write.

5 Perfect Squares, Written Three Ways

We said earlier that some integers are squares of other integers and some are not; for those that are not, the square roots are irrational — we proved, for instance, that $\sqrt{2}$ is irrational. A perfect square is an integer whose square root is also an integer:

$$\{ z \mid z \in \mathbb{Z},\ \sqrt{z} \in \mathbb{Z} \}$$

Now, a square must be non-negative, as we have already discussed: multiplying two negative numbers gives a positive number, and multiplying two positive numbers again gives a positive number. A perfect square is therefore always non-negative — it could be 0. So we may as well let the **generator** be the natural numbers instead:

$$\{ m \mid m \in \mathbb{N},\ \sqrt{m} \in \mathbb{N} \}$$

Only the positive root is needed

The number 4 has two square roots: $4 = (-2) \times (-2)$ and $4 = 2 \times 2$. But it is enough to know that *one* of its square roots is an integer, since the other is just the same with a minus sign.

Turning the filter into a transformation

We can go further and replace the filter by a transformation. Every natural number, when squared, gives a natural number, and all the perfect squares arise in that way. So instead of looking for numbers whose square root is a natural number, we can simply take every natural number and square it:

$$\{ m^2 \mid m \in \mathbb{N} \} = \{ 0,\ 1^2,\ 2^2,\ 3^2,\ \ldots \}$$

Three descriptions of the same set

$$\{ z \mid z \in \mathbb{Z},\ \sqrt{z} \in \mathbb{Z} \} = \{ m \mid m \in \mathbb{N},\ \sqrt{m} \in \mathbb{N} \} = \{ m^2 \mid m \in \mathbb{N} \}$$

In the first step we changed the *generating set* from $\mathbb{Z}$ to $\mathbb{N}$, using a property of perfect squares. In the second we converted the *filter* into a *transformation*: rather than putting a condition on the numbers generated, we took all of them and squared them.

6 Perfect Squares Among the Rationals

We can extend the notion of a perfect square to other families of numbers. A rational is a perfect square if it is the square of another rational. Since integers can now appear both above and below the line:

Example — $\frac{9}{16}$ and $\frac{1}{2}$

$$\frac{9}{16} = \frac{3^2}{4^2} = \frac{3}{4} \times \frac{3}{4}$$ — a perfect square in the world of rationals

Not everything qualifies, though. Since $\sqrt{2}$ cannot be represented as a rational, it is easy to check that $\frac{1}{2}$ cannot be written in the form $\left(\frac{p}{q}\right)^2$. So some rationals are perfect squares and some are not.

Changing $\mathbb{Z}$ and $\mathbb{N}$ to $\mathbb{Q}$ in the definitions above gives a perfectly reasonable notion of perfect squares in this new domain:

$$\{\, q \mid q \in \mathbb{Q},\ \sqrt{q} \in \mathbb{Q} \,\} \qquad \text{or, in the second form,} \qquad \{\, q^2 \mid q \in \mathbb{Q} \,\}$$

The generator determines the set

Depending on how the generator is chosen, we may end up with the same set or with a genuinely different one. It is therefore important to specify all the parts of a set comprehension correctly, so that there is no ambiguity and we obtain exactly the set we mean to obtain.

Lecture 7 — Counting with Venn Diagrams

Union, intersection, difference and complement revisited; solving numerical problems about sets

What this lecture covers

Venn diagrams are not just pretty pictures — they are a genuinely useful way of reasoning about sets. After recalling the four operations, we use them to solve a family of counting problems: given how many students take each subject, how many take both, and how many take neither, we can recover whichever quantity is missing.

1 The Operations, Recalled

Union and intersection

The **union** takes two sets, combines them and removes the duplicates. The overlapping part of the diagram holds the common elements:

$$\{a, b, c\} \cup \{c, d, e\} = \{a, b, c, d, e\}$$

Here c sits in the overlap, a and b belong only to X, and d and e only to Y.
Conversely, the **intersection** keeps only what is common to the two:

$$\{a, b, c, d\} \cap \{a, d, e, f\} = \{a, d\}$$

with b and c only on the left and e and f only on the right.

Set difference tells us what is on the left but not on the right. And the **complement** can be taken once we have an overall **universe** to talk about: with respect to that set, we ask which elements are *not* in the set we are looking at.

The universe matters

If we take the natural numbers as a whole, the primes are a subset of $\mathbb{N}$, and the complement of the primes is all those natural numbers that are not prime. But if we took the complement of the primes with respect to $\mathbb{R}$, we would pick up all sorts of numbers that are not even integers. So whenever we define a complement we must define the universe we are working in.

2 A First Counting Problem

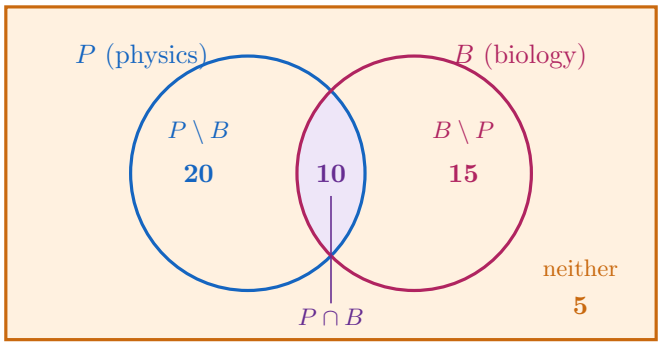
Problem 1 — how big is the class?

In a class, 30 students have taken physics and 25 have taken biology. Of these, 10 have taken *both*, and there are 5 who have taken *neither*. How many students are there in the class?

We represent the two groups of students — those taking physics and those taking biology — by two sets P and B. Since some take both, the two sets overlap.

- The overlap has 10 students, so we write 10 in the intersection: $|P \cap B| = 10$.
- 30 students took physics overall, and 10 of them are already accounted for in the overlap. So 20 students took physics but not biology: $|P \setminus B| = 30 - 10 = 20$.

- Symmetrically, 25 took biology and 10 of them took both, so $|B \setminus P| = 25 - 10 = 15$.
- Finally, 5 students took neither subject, so they lie outside $P \cup B$ — in the complement.



There are four regions of interest, and they are all **disjoint** from one another: physics but not biology, biology but not physics, both, and neither. Adding the students across them gives the exact class strength:

$$|P \setminus B| \;+\; |P \cap B| \;+\; |B \setminus P| \;+\; |\text{neither}| \;=\; 20 + 10 + 15 + 5 \;=\; \mathbf{50}$$

Reading the answer

The subject enrolments add up to $30 + 25 = 55$, yet the total class strength is only 50 — the 10 students taking both have been counted twice. And in fact only 45 students are taking these subjects at all, since 5 have taken neither.

3 The Same Problem, Data Given Differently

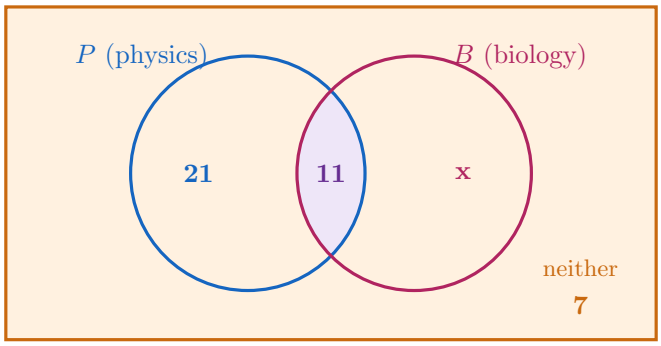
Problem 2 — how many took biology but not physics?

The class strength is 55. Of these, 32 students took physics, 11 took both physics and biology, and 7 took neither. How many took biology but not physics?

Again we draw the Venn diagram. As before, 11 goes in the intersection, since that is the number who took both. Since 32 took physics, subtracting the 11 gives

$$|P \setminus B| \;=\; 32 - 11 \;=\; 21,$$

and the complement holds 7. Call the unknown region $x = |B \setminus P|$.



total class strength = 55

Since the four disjoint regions must add up to the total:

$$7 + 21 + 11 + x \;=\; 55 \qquad \Longrightarrow \qquad x \;=\; 55 - 39 \;=\; \mathbf{16}$$

So 16 students have taken biology but not physics.

4 Finding the Overlap

Problem 3 — how many took both?

There are 60 students in the class. Of these, 35 took physics, 30 took biology, and 10 took neither. How many took both subjects?

Here we use the notation introduced when we first met sets: a pair of vertical bars around a set denotes its **cardinality**, that is, the number of elements in it.

Cardinality notation

$$|Y| \;=\; \text{the number of elements in the set } Y$$

We are told that

$$|P| = 35, \qquad |B| = 30, \qquad \text{so} \qquad |P| + |B| = 35 + 30 = 65.$$

This counts 65 enrolments in physics or biology or both. But there are 60 students in the class, of whom 10 took neither, so the union actually contains only

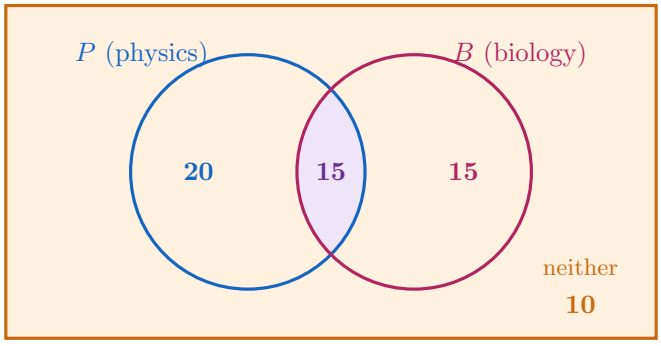
$$|P \cup B| \;=\; 60 - 10 \;=\; 50$$

students. So 65 enrolments are spread over only 50 students: some of them must be taking both subjects and are therefore being counted twice. The number counted twice must be the difference:

$$|P \cap B| \;=\; \big(|P| + |B|\big) \;-\; |P \cup B| \;=\; 65 - 50 \;=\; \mathbf{15}$$

Equivalently, rearranged:

$$|P \cup B| \;=\; |P| \;+\; |B| \;-\; |P \cap B|.$$



$35 - 15 = 20, \quad 30 - 15 = 15, \quad \text{total} = 20 + 15 + 15 + 10 = 60$

The intersection was obtained by taking the class total, setting aside the 10 who took neither, and computing the difference between the number of subject enrolments and the number of students actually registered for one or both subjects.

Drawing the right diagram

Venn diagrams are not simple doodles: they are a practical tool for calculating properties of sets, especially in numerical problems. What matters is being able to draw the *proper* diagram — which groups overlap, how they overlap, and which regions are empty — and then to fill the disjoint regions with numbers.

Lecture 8 — Relations

Cartesian products, binary and n-ary relations, reflexivity, symmetry, transitivity and equivalence classes

What this lecture covers

Having seen sets, we now move on to **relations**. We first meet a new way of combining sets — the Cartesian product, which pairs up elements — and then carve subsets out of that product to obtain relations. We look at relations on numbers, on people and on teachers and courses, extend from pairs to n-tuples, and finish with the properties a relation may have: reflexivity, symmetry, transitivity, anti-symmetry, and their combination into an equivalence relation.

1 Where We Have Reached

A set is a collection of items, and we can construct new sets from old ones: unions combine two sets into one, intersections take the common elements, the difference takes the elements of X not in Y, and — once a universe is fixed — the complement takes those elements not in X.

To carve out subsets we use set comprehension:

Three comprehensions we have met

$$\{\, x \mid x \in \mathbb{Z},\ x \bmod 2 = 1 \,\} \qquad \text{the odd integers}$$

$$\left\{ \frac{p}{q} \,\middle|\, \frac{p}{q} \in \mathbb{Q},\ \gcd(p,q) = 1 \right\} \qquad \text{fractions in reduced form}$$

$$[3, 17) \;=\; \{\, r \mid r \in \mathbb{R},\ 3 \le r < 17 \,\} \qquad \text{left closed, right open}$$

2 The Cartesian Product

Cartesian product

The **Cartesian product** of two sets takes one element from each and forms a pair:

$$A \times B \;=\; \{\, (a, b) \mid a \in A,\ b \in B \,\}$$

The first element of the pair comes from A and the second from B.

Example — a small product

With $A = \{0, 1\}$ and $B = \{2, 3\}$:

$$A \times B \;=\; \{\, (0,2),\ (0,3),\ (1,2),\ (1,3) \,\}$$

four possible pairs in all.

Order matters in a pair

In a set the order of the elements is not important, but in a pair the left element comes from

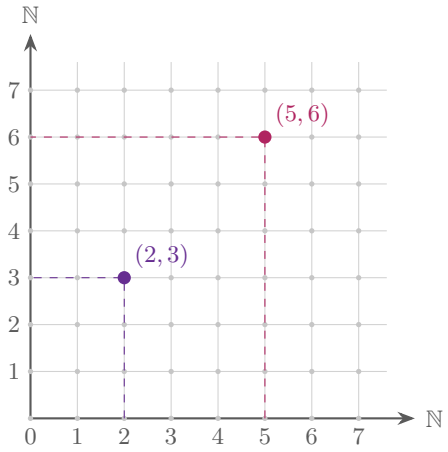
the left part of the product and the right element from the right part. So

$$(0,1) \neq (1,0).$$

We must respect the order when we talk about a pair.

Visualising the product

For sets of numbers we normally picture the product as a space, drawn as a familiar graph. For $\mathbb{N} \times \mathbb{N}$ we draw a grid, with one copy of $\mathbb{N}$ on the x-axis and another on the y-axis. The pair $(2,3)$ is the point with x-coordinate 2 and y-coordinate 3.



Take the first coordinate on the x-axis and the second on the y-axis; where the two meet in the grid is the point we want. The same picture works for the reals, except that the plotted points then have real coordinates rather than only natural-number ones.

3 Relations

The Cartesian product consists of *all* possible pairs. Just as with set comprehension, we often want to pick out only some of them — and that is a **relation**.

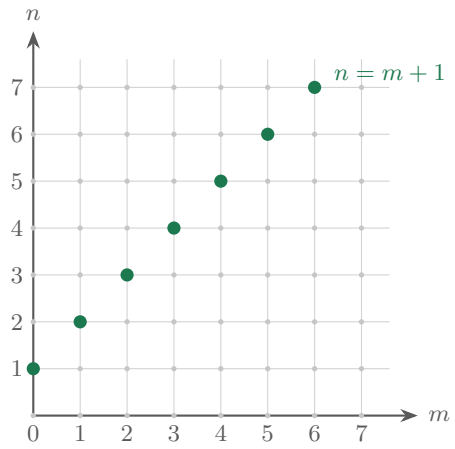
Binary relation

A **binary relation** is a *subset of the product*: we take all possible pairs and apply a condition that filters out the pairs of interest.

Example — the successor relation

$$\{\,(m,n) \mid (m,n) \in \mathbb{N} \times \mathbb{N},\ n = m+1\,\}$$

This contains $(0,1)$ since $1 = 0+1$; $(2,3)$ since $3 = 2+1$; $(17,18)$; and so on.



Only a subset of the grid points survives — exactly those satisfying the condition.

Example — the divisibility relation

$$\{\,(d,n) \mid (d,n) \in \mathbb{N} \times \mathbb{N},\ d \mid n\,\}$$

Recall that $d \mid n$ means $n \bmod d = 0$. So $(2, 82)$ is in the relation, since 2 is a factor of 82, and so is $(14, 56)$, since 14 is a factor of 56.

Two notations for the same thing

To say that a pair belongs to a relation R we may either treat R as a set and write

$$(a,b) \in R,$$

or treat R as an operator and write

$$a\,R\,b \qquad \text{“}a \text{ is related by } R \text{ to } b\text{”}.$$

Different books use different conventions; they mean exactly the same thing.

4 Relations Beyond Numbers

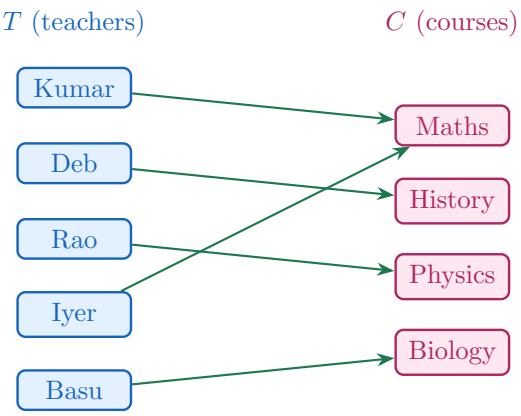
Example — teachers and courses

Let T be the set of teachers in a school and C the set of courses offered this term. We need to describe which teachers teach which courses, so we define an **allocation relation**:

$$A \;=\; \{\,(t,c) \mid (t,c) \in T \times C,\ t \text{ teaches } c\,\}$$

In principle every teacher could teach every course, but of course that is not the case: some teachers teach some courses, and we keep precisely those pairs.

Here is a different graphical way of drawing a relation — not the grid, but a picture with **nodes** for the elements and **arrows** joining the pairs in the relation.



We read off that Kumar teaches maths, Deb teaches history, and so on. This is a useful way of visualising relations on finite sets, and we shall see it often.

Example — mothers and children

Let P be the set of people in a country. From all pairs of people we isolate those in which the first element is the mother of the second:

$$\{\,(m,c) \mid (m,c) \in P \times P,\ m \text{ is the mother of } c\,\}$$

5 A Relation That Draws a Circle

Suppose we want all the points of $\mathbb{R} \times \mathbb{R}$ lying at distance 5 from the **origin** $(0,0)$. The distance of a point (a,b) from the origin is given by Pythagoras' theorem:

$$\text{distance} \;=\; \sqrt{a^2+b^2}$$

So the relation we want is

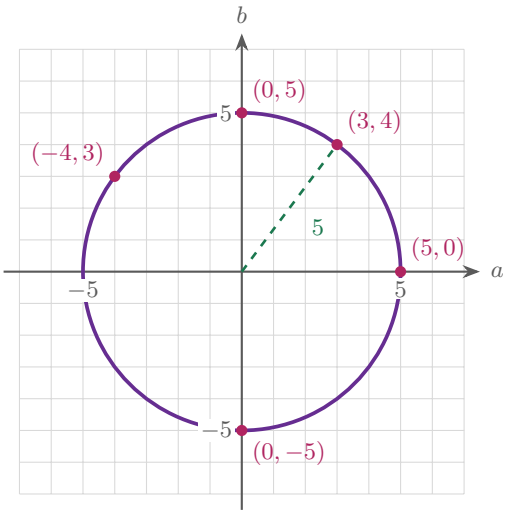
$$\left\{\,(a,b) \;\middle|\; (a,b) \in \mathbb{R}\times\mathbb{R},\ \sqrt{a^2+b^2}=5\,\right\}$$

Some points on it

$$(0,5): \quad \sqrt{0+25} \;=\; 5 \qquad \checkmark$$

$$(3,4): \quad 3^2=9,\ \ 4^2=16,\ \ 9+16=25,\ \ \sqrt{25} \;=\; 5 \qquad \checkmark$$

Plotting every such point gives a **circle of radius 5 centred at the origin**.



So relations can define interesting geometric shapes, and we often deal with geometric shapes in this relational form because it is easier to manipulate than a picture.

Reduced rationals, seen as a relation

We described rationals in reduced form as p/q with $\gcd(p, q) = 1$. But since every rational is really a pair of integers — the numerator and the denominator — we may equally think of this as a relation on the integers:

$$\{\,(p, q) \mid (p, q) \in \mathbb{Z} \times \mathbb{Z},\ \gcd(p, q) = 1\,\}$$

6 From Pairs to n-Tuples

We need not restrict ourselves to binary relations: the Cartesian product notation extends to any number of sets.

Example — Pythagorean triples

Pythagoras' theorem says that the square on the hypotenuse is the sum of the squares on the other two sides. Which values of a, b and c can be the sides of a right triangle?

$$\{\,(a, b, c) \mid (a, b, c) \in \mathbb{N} \times \mathbb{N} \times \mathbb{N},\ a, b, c \neq 0,\ a^2 + b^2 = c^2\,\}$$

Here we have three copies of $\mathbb{N}$. The sides must all be non-zero — we do not want triangles in which one side has collapsed to a point.

Example — squares in the plane

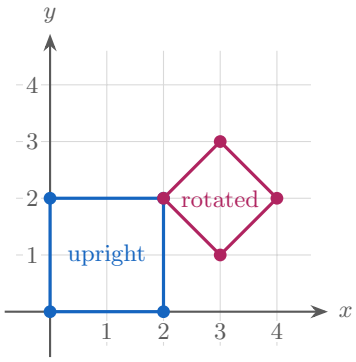
A corner of a square is a point $(x, y) \in \mathbb{R} \times \mathbb{R}$, and we want four corners which, joined by lines, form a square. For instance the four points

$$(0, 0), \quad (0, 2), \quad (2, 0), \quad (2, 2)$$

form a square; and a set of points forming a rotated square — a diamond — is a square too, since rotating it upright shows that it is.
There are many such four-element sets of points. So "square", viewed as a relation, is a relation on

$$\mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R}^2$$

— one $\mathbb{R}^2$ per corner — which is really a relation on **eight** copies of $\mathbb{R}$, or, if we prefer to group them, four copies of pairs of $\mathbb{R}$.



So we can take relations on an arbitrary number of copies of a set: from pairs we move to triples, to quadruples, and in general, with n copies, to an **n-tuple**.

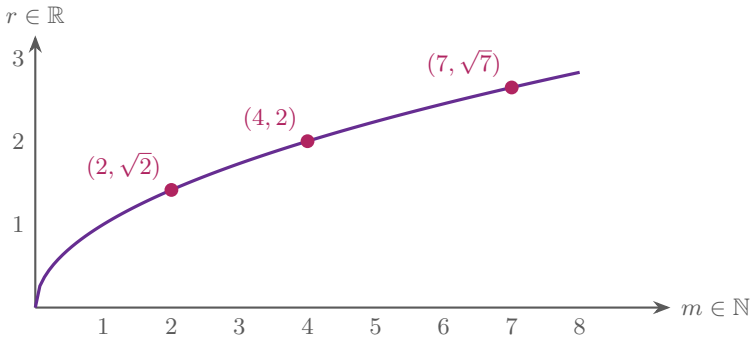
$$X_1 \times X_2 \times \cdots \times X_n \;=\; \{\,(x_1, x_2, \ldots, x_n) \;\mid\; x_1 \in X_1,\; x_2 \in X_2,\; \ldots,\; x_n \in X_n\,\}$$

The sets $X_1, \ldots, X_n$ may be different or the same; we take one element from each and form an n-tuple. Picking out a particular subset of these n-tuples gives a relation.

Example — square roots across two different sets

$$\{\,(m, r) \;\mid\; (m, r) \in \mathbb{N} \times \mathbb{R},\; r = \sqrt{m}\,\}$$

This contains $(2, \sqrt{2})$, $(4, 2)$, $(7, \sqrt{7})$, and so on.



A note on the picture

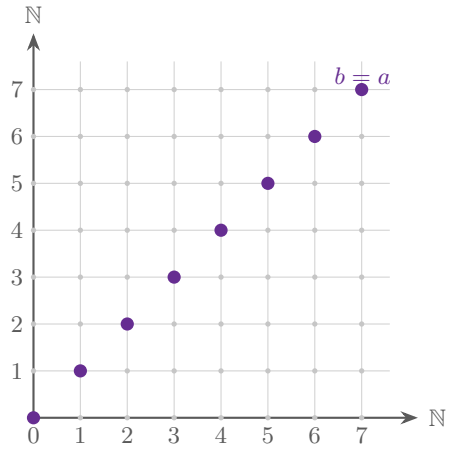
The y-axis here is elongated compared with the x-axis, so the plot is not to the same scale in both dimensions — that is simply how the square-root function behaves.

7 The Identity Relation

Some binary relations turn up all over the place, so it is useful to know their names. The first is the **identity relation**, which maps every element to itself. Since identity means equality, it is defined on two copies of the *same* set:

$$\{\,(a, b) \;\mid\; (a, b) \in A \times A,\; a = b\,\}$$

In other words we want pairs of the form (a, a). Plotted on $\mathbb{N} \times \mathbb{N}$ this gives $(0, 0), (1, 1), (2, 2), \ldots$ — the diagonal of the grid.



Shorthand notations

Writing the full comprehension is sometimes tedious, so one often sees the abbreviations

$$\{\,(a,a) \mid (a,a) \in A \times A\,\} \qquad \text{or even} \qquad \{\,(a,a) \mid a \in A\,\},$$

where in the last form we pull one copy of the element out of A and build a pair from two copies of it. All of these are equivalent ways of writing the same thing, although only the first form technically follows the notation we introduced for relations.

8 Properties of Relations

Reflexivity

Reflexive

A relation R is **reflexive** if every element is related to itself:

$$\text{for every } a: \quad (a,a) \in R.$$

Equivalently, the identity relation is *included* in R.

This does not mean the identity pairs are the only ones. The identity relation contains *only* the reflexive pairs; a reflexive relation must contain all of them, but may contain others too.

Example — divisibility is reflexive

Provided we exclude 0, divisibility is reflexive, since every number divides itself:

$$a \mid a \qquad \text{for every } a \neq 0.$$

Symmetry

Symmetric

A relation is **symmetric** if

$$(a,b) \in R \quad \Longrightarrow \quad (b,a) \in R.$$

Example — coprimality

Take the pairs whose greatest common divisor is 1, that is, those with no common divisor — as happens with reduced fractions. It does not matter whether we write (a, b) or (b, a): if $\gcd(a, b) = 1$ then $\gcd(b, a) = 1$. So both pairs are in the relation, or neither is.

Example — differing by 2

$$\{\,(a, b) \mid |a - b| = 2\,\}$$

The absolute value takes the difference and removes the negative sign. If $(5, 7)$ is in the relation then so is $(7, 5)$, since both have the same difference in absolute value. Ordinarily subtraction gives a sign difference, but because we take the absolute value there is none here. So this relation is symmetric.

Transitivity

Transitive

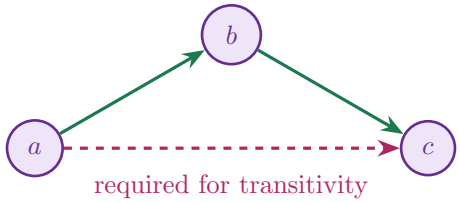
A relation is **transitive** if, whenever two related pairs share an element, the ends are related too:

$$(a, b) \in R \text{ and } (b, c) \in R \implies (a, c) \in R.$$

Examples of transitivity

$$2 \mid 6 \text{ and } 6 \mid 36 \implies 2 \mid 36$$

$$3 < 10 \text{ and } 10 < 28 \implies 3 < 28$$



Anti-symmetry

Anti-symmetric

Symmetry says that if $(a, b) \in R$ then $(b, a) \in R$. **Anti-symmetry** says the opposite:

$$(a, b) \in R \implies (b, a) \notin R.$$

Examples of anti-symmetry

Strictly less than. If $a < b$ then it cannot be that $b < a$. (Note that "<" is transitive as well, as we just saw.)
Mothers and children. If p is the mother of c, then c certainly cannot be the mother of p.

What anti-symmetry does not require

Anti-symmetry does not insist that one of the two pairs must be present. There may well be p and c such that neither is the mother of the other — that is allowed. It says only that *if* a pair is related one way, it must not be related the other way.

9 Equivalence Relations

Combining three of these conditions gives an especially interesting class of relations.

Equivalence relation

A relation is an **equivalence relation** if it is

reflexive, **symmetric**, and **transitive**.

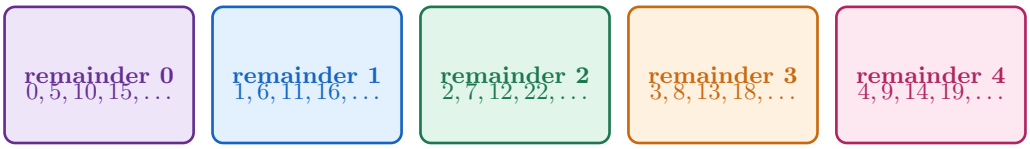
Example — the same remainder modulo 5

Connect together all numbers having the same remainder modulo 5. For instance 7 leaves remainder 2 on division by 5, and so does 22, so 7 and 22 are related.
If two numbers have the same remainder modulo 5, then going from one to the other moves in multiples of 5: indeed $22 - 7 = 15$. So in set notation,

$$\{\,(a,b) \mid (a,b) \in \mathbb{Z} \times \mathbb{Z},\ (b-a) \bmod 5 = 0\,\}$$

We are not asking what the actual remainders of a and b are; we are saying only that they are separated by a multiple of 5, and hence must leave the same remainder.

This divides the integers into **five** groups according to the remainder: those divisible by 5 (remainder 0); those like 6 and 11 (remainder 1); those like 7 and 12 (remainder 2); and so on. Since there are five possible remainders $0, 1, 2, 3, 4$, the integers split into five disjoint classes.



Example — the clock

A familiar case of modulo arithmetic is an ordinary clock. It measures time from 0 to 12 and then cycles around again. So although there are 24 hours in a day, the clock partitions them into classes in which 0 and 12 are the same, 1 and 13 are the same, and so on — there is no distinction on the face between 2 a.m. and 2 p.m.
The clock is showing us an equivalence class of hours, treating a.m. and p.m. as equal, and we have to know from context which is meant.

Equivalence classes

The main thing to note is that an equivalence relation **partitions** a set into disjoint groups: all the elements within a group are equivalent to one another, and elements in different groups are not. These groups are called **equivalence classes**.

Equivalence classes represent a kind of equality. We are often happy to work with this notion of equality rather than with actual equality, since it has very much the same properties.

Lecture 9 — Functions

Domain, codomain and range; the relation behind a function; injective, surjective and bijective functions

What this lecture covers

Closely related to relations are **functions**. We define a function as a rule that converts an input into an output, fix the three sets that go with it — domain, codomain and range — and see how every function carries a binary relation with it. We then study the properties that make functions useful: injectivity, surjectivity and their combination, bijectivity, which turns out to be the right tool for comparing the sizes of sets.

1 What Is a Function?

A **function** is a rule telling us how to convert an input into an output. Suppose we want a function that, given x, returns x^2. One way to write the rule is

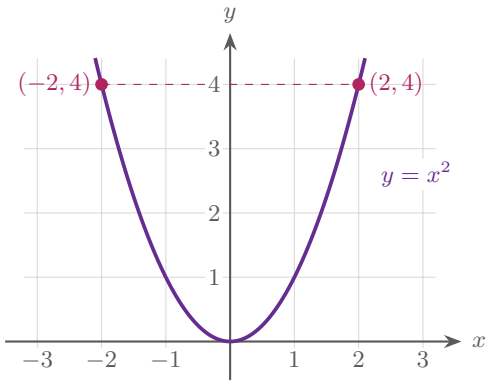
$$x \mapsto x^2$$

which says: given an x, it is transformed into x^2. More conventionally we also give the function a name:

$$\text{sq}(x) = x^2$$

So sq takes a parameter x as input and produces an output that transforms that parameter — here, its square.

We can plot x against x^2 by marking all the points whose second coordinate is the function value of the first. This gives the familiar parabola.



Notice the symmetry about the y-axis: since 2^2 is the same as $(-2)^2$, and 3^2 the same as $(-3)^2$, and so on, the two halves mirror each other.

2 Domain, Codomain and Range

When we define a function we must be careful to specify what set the input is taken from and what set the output lands in.

The three sets

- **Domain** — the input set. For sq as defined above the domain is $\mathbb{R}$: we can square any real number.
- **Codomain** — the set of possible output values. For sq this is also $\mathbb{R}$.
- **Range** — the values the function can *actually* take. The range is a **subset** of the codomain.

Even if the input is negative, the square of a number is positive, so we cannot obtain all the reals as outputs of sq. Its range is the non-negative reals:

$$\mathbb{R}_{\geq 0} \;=\; \{\, r \;\mid\; r \in \mathbb{R},\; r \geq 0 \,\}$$

To specify a function abstractly, together with its domain and codomain, we write

$$f : X \longrightarrow Y$$

which tells us — without saying what the function actually does — on what sets it operates: X is the input set and Y the output set.

3 The Relation Behind a Function

The close connection between functions and relations is that with every function f we can associate a relation R_f, consisting of all the input–output pairs the function allows.

$$R_{\text{sq}} \;=\; \{\, (x, y) \;\mid\; y = x^2 \,\}$$

This is often simplified by writing just

$$y \;=\; x^2,$$

rather than naming $f(x)$ and then saying $f(x) = y$. This is an implicit notation in which we implicitly name the output for each x as y.

Since a relation is a subset of a Cartesian product, the product here is formed from the domain and the range:

$$R_f \;\subseteq\; \text{domain} \times \text{range}.$$

The two defining properties of R_f

Totality. Defining the domain means the function is defined at *every* value in it:

$$\text{for every } x \in \text{domain}(f) \quad \text{there is a } y \text{ with } (x, y) \in R_f.$$

Uniqueness. A function is a rule for producing an output, so there can be no confusion about what that output is:

$$\text{for each } x \quad \text{there is exactly } \textbf{one } y \text{ with } (x, y) \in R_f.$$

Graphs of functions are graphs of relations

When we draw the graph of a function — the parabola above, for instance — we are really plotting all the points satisfying the relation R_f. So plotting a graph is the same activity for functions and for relations, because implicitly we are always plotting the relation corresponding to the function.

4 Linear Functions

A function of the form $mx + c$ defines a **line**. For example:

$$f(x) \;=\; 3.5\,x + 5.7$$

The quantity multiplying x is the **slope**: it determines the angle at which the line goes. The quantity without x is the **intercept**. Setting $x = 0$ makes the first term vanish:

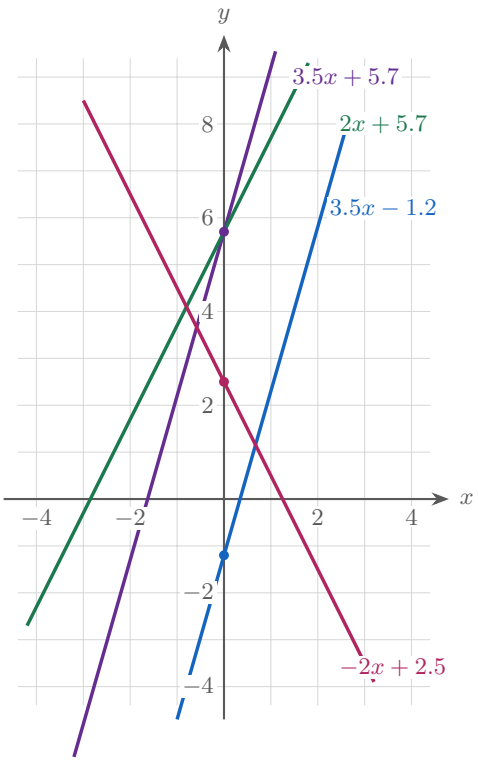
$$f(0) \;=\; 3.5 \times 0 + 5.7 \;=\; 5.7,$$

so the second term tells us where the line crosses the y-axis.

Changing these two values gives different lines.

A family of lines

$y = 3.5\,x + 5.7$	the original line
$y = 3.5\,x - 1.2$	same slope — parallel, but crossing the y-axis lower
$y = 2\,x + 5.7$	same intercept, shallower slope (reduced from 3.5 to 2)
$y = -2\,x + 2.5$	a negative slope, so the line comes down; intercept 2.5



For all these functions the domain is $\mathbb{R}$ and the codomain is $\mathbb{R}$. And since a line runs from far down at $-\infty$ to far up at $+\infty$ — whether it rises or falls — it takes every real value. So here the codomain and the **range** are both equal to $\mathbb{R}$.

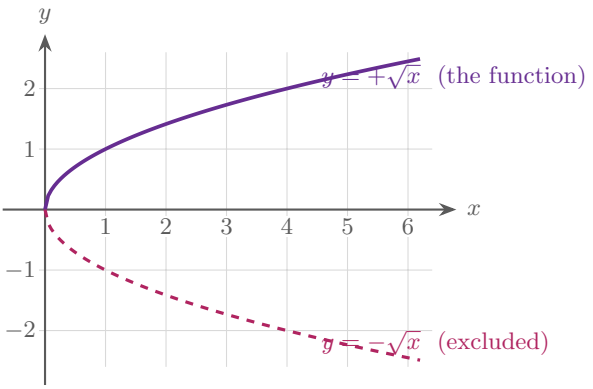
5 The Square Root Function

Consider $x \mapsto \sqrt{x}$. The first question is: *is this a function?* Recall that a function must be defined on every input value and must have a unique output.

Squaring a negative number gives the same result as squaring the positive one: $5^2 = (-5)^2 = 25$. So on taking $\sqrt{25}$ we cannot tell whether we mean $+5$ or -5.

The positive-root convention

When we write $\sqrt{x}$ as a function, our convention is that we take the **positive** square root. The negative root would give a symmetric curve below the axis; taking both together would *not* be a function, since each x would then have two possible outputs, which is not allowed.



What is the domain?

That depends on what we allow the codomain to be. No real number multiplied by itself gives a negative answer, by the law of signs. So if we insist that the output be a real number, the function is defined only for non-negative inputs:

$$\text{domain} \;=\; \mathbb{R}_{\geq 0} \;=\; \{\, r \mid r \in \mathbb{R},\; r \geq 0 \,\}.$$

On the other hand, the complex numbers include $\sqrt{-1}$ and hence the square roots of all negative numbers. Once we allow complex numbers as outputs, square root can be defined on *all* of $\mathbb{R}$.

Domain and range are context-dependent

The notions of domain and range are flexible, depending on how we intend to use the function. We must therefore always be clear about the context in which a function is being used.

6 Injective Functions (One-to-One)

Just as relations had interesting properties, so do functions. The first is whether the function is **one-to-one**, or **injective**: do different inputs always produce different outputs?

Injective

$$x_1 \neq x_2 \quad \Longrightarrow \quad f(x_1) \neq f(x_2)$$

A line is injective; a parabola is not

For the linear function seen above, changing x moves us along the line to a new point, so no two x values point to the same y. The line is **injective**.
For a parabola such as $f(x) = 7x^2$, we have already seen that

$$f(a) \;=\; f(-a),$$

so the plus and minus versions give the same output. Distinct inputs do not produce distinct outputs, and the square function is **not injective**.

7 Surjective Functions (Onto)

We distinguished the codomain — the set into which the function produces answers — from the range, the values the function actually takes. The question now is whether all the values in the codomain are actually reached. This is **surjectivity**, or being **onto**.

Surjective

The range of a surjective function equals its codomain:

$$\text{for every } y \in \text{codomain}(f) \quad \text{there is an } x \in \text{domain}(f) \text{ with } f(x) = y.$$

A line is surjective

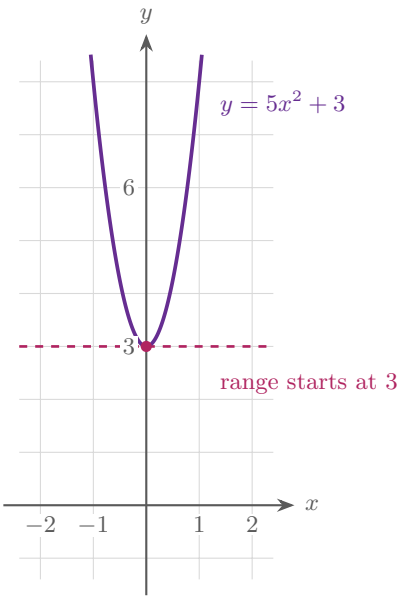
For a line, pick any point y; we can solve for x and find a value that gives that y. So a linear function is surjective.

A shifted parabola is not

Take $f(x) = 5x^2 + 3$. Without the $+3$ term the parabola could take only non-negative values, since x^2 is always non-negative. Adding 3 means it can take only values 3 and above:

$$\text{codomain} = \mathbb{R}, \qquad \text{range} = \{\, r \in \mathbb{R} \mid r \geq 3 \,\}$$

so this is definitely not surjective.



Nor is $7\sqrt{x}$

Even taking the codomain to be $\mathbb{R}$, we know that the square root of a non-negative number never gives a negative answer, because by convention we take the positive root. So $7\sqrt{x}$ is not surjective either.

8 Bijections

Combining the two properties gives a **bijective** function.

Bijective

A function is **bijective** if there is a **one-to-one correspondence** between the domain and the codomain: every x in the domain maps to a distinct y in the codomain, and every y in the codomain has a unique x mapping to it.

$$f \text{ is bijective} \iff f \text{ is injective } \textbf{and} \text{ surjective}$$

This may look obvious, but only one direction really is.

The argument

Bijective ⇒ injective and surjective. This is immediate from the definition. It says every x maps to a distinct y, so no two x's map to the same y — that is injectivity. And it says every y in the codomain has a unique pre-image — that is surjectivity.
Injective and surjective ⇒ bijective. We must guarantee, first, that every x maps to a unique y: this holds because injectivity says that if $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$.
Next, surjectivity says everything in the output comes from some input — but not necessarily a unique one. Suppose some y had two pre-images x_1 and x_2 mapping to it. Then those two pre-images would violate injectivity. So combining surjectivity with injectivity, the pre-image must be unique, and the two conditions together guarantee a bijection.

9 Counting with Bijections

An important use of a bijection is to **count** the items in a set. Recall that the cardinality of a set is the number of items in it, and that for a finite set we can count them.

The two sacks of marbles

Suppose somebody gives you two large sacks filled with marbles and asks whether the two sacks have the same number. Think of the sacks as sets and the marbles as a large number of elements.
You could count the marbles in each sack, but this is tedious: keeping track of many small objects, we often lose count, miscount, or add one too many. To be sure, we would have to count several times over — an error-prone process if done by hand.
Here is a less error-prone manual method. Put a hand into each sack, pull out one marble from each, and set them aside. Repeat: one marble from each sack every time.

- If both sacks become empty *together*, then we pulled out one from each every time, so we have established a one-to-one correspondence between the marbles in the first sack and those in the second.
- If one sack is empty while the other is not, then up to that point we pulled an equal number from both, and now one sack has extra marbles — so they were not equal.



one-to-one correspondence ⇒ same cardinality

Bijection means equal cardinality

Establishing a bijection is equivalent to saying that two sets have the same cardinality. For finite sets this is a convenience; for **infinite** sets it is the *only* way to establish that the cardinalities are the same.

10 Counting the Lines in a Plane

Suppose we want to know whether the number of lines we can draw is the same as the number of points in the plane $\mathbb{R} \times \mathbb{R}$. It may not be obvious how to argue this either way.

But recall that every line can be written as $mx + c$, and that changing m gives a new line, as does changing c. So m and c *together* uniquely define a line:

$$(m, c) \longleftrightarrow \text{the line } y = mx + c$$

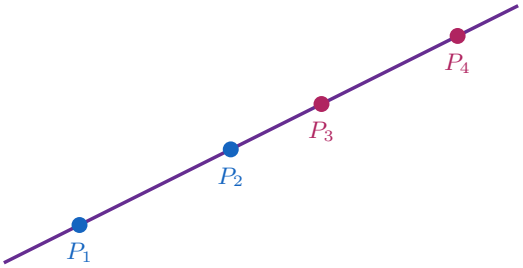
Every pair (m, c) defines a line, and every line defines a pair (m, c). That is a bijection between the lines and the points of the plane, so the number of lines we can draw on a plane is the same as $|\mathbb{R} \times \mathbb{R}|$ — the number of points on a plane. This is worth thinking about, since it is not obvious at first sight.

Where the argument breaks down

Now let us try to extend it. From geometry we know that through any two points there passes a **unique** line. We have just claimed that the set of lines has the same cardinality as $\mathbb{R} \times \mathbb{R}$. Since every pair of points defines a line, can we conclude that the set of *pairs of points* — that is, $\mathbb{R}^2 \times \mathbb{R}^2$ — has the same cardinality as $\mathbb{R} \times \mathbb{R}$?

The bijection fails

The problem is that this is **not** a bijection. Along any line there are many points. Two given points determine a unique line, but we get the *same* line from a different pair of points on it. So it is not the case that every pair of points generates a different line.



the pair (P_1, P_2) and the pair (P_3, P_4) give the *same* line
— so pairs of points do not correspond one-to-one with lines

Unless we can show that different pairs of points generate different lines, we do not have a one-to-one correspondence between pairs of points and lines, and the argument collapses.

The moral

Whenever we use a bijection to describe a correspondence — especially to count the elements of an infinite set, or to compare infinite sets against each other — we must make sure that the function we are defining really *is* a bijection.

Lecture 10 — Relations, Tables and Joins

Cartesian products revisited; relations beyond numbers; why every table is a relation

What this lecture covers

We defined relations as subsets of a Cartesian product with special properties. Here we look at relations again and ask why we care about them so much. After working carefully through products and several number relations, we move beyond numbers to an airline's route map and then to tables of data — because, from a computing and data science point of view, **a table is a relation**, and combining tables is an operation on relations.

1 Cartesian Products, Carefully

A Cartesian product takes all pairs of elements from a collection of sets. In $A \times B$ we take every pair (a, b) with a from A and b from B; the order matters, the first element coming from the first set and the second from the second.

Example — a concrete product

Let $A = \{1, 4, 7\}$ and $B = \{1, 16, 49\}$. Then

$$\begin{aligned} A \times B \;=\; \{\,&(1,1),\ (1,16),\ (1,49),\\ &(4,1),\ (4,16),\ (4,49),\\ &(7,1),\ (7,16),\ (7,49)\,\} \end{aligned}$$

We take 1 and combine it with 1, 16 and 49; then 4 with each of them; then 7.

$$|A| = m, \quad |B| = n \qquad \Longrightarrow \qquad |A \times B| \;=\; m \times n$$

Each of the m elements is paired with each of the n elements, giving $m \times n$ pairs.

$A \backslash B$	1	16	49
1	(1, 1)	(1, 16)	(1, 49)
4	(4, 1)	(4, 16)	(4, 49)
7	(7, 1)	(7, 16)	(7, 49)

Two things to remember

The product is ordered. Reversing it does not give the same set of pairs — every pair is reversed:

$$B \times A \;\ni\; (16,1),\ (49,1) \qquad \text{in place of} \qquad (1,16),\ (1,49)$$

There is no constraint on what we may multiply. We can take the product of a set with itself, and $B \times B$ contains not only the identical pairs but the dissimilar ones too:

$$B \times B \;\ni\; (1,1),\ (16,16),\ (49,49),\ (1,16),\ (16,49),\ (49,16),\ \ldots$$

More than two sets

Nothing restricts us to two sets: in general a Cartesian product takes any number of sets and gives **tuples**. With three sets each element of the product has three components, in order.

Example — $A \times B \times A$

Taking every element of A, then of B, then of A again:

$$(1,1,1),\quad (1,1,4),\quad (1,1,7),\quad (1,16,1),\quad (1,16,4),\quad (1,16,7),\quad \ldots$$

The second copy of A and the first copy of A are different positions, even though they are the same set.

Order of listing versus order within a tuple

The Cartesian product is ultimately a *set*, so it does not matter in what order we write the triples down. But it is convenient to write them systematically, running through each set one by one — otherwise we may miss something.

2 From Products to Relations

Cartesian products are the **building blocks** of relations. What we finally want is not all the pairs or triples, but the ones of interest to us.

Example — picking three pairs out of nine

From $A \times B$ we might be interested only in the pairs where each element of A is matched with the element in the corresponding position of B:

$$S \;=\; \{\,(1,1),\ (4,16),\ (7,49)\,\} \;\subseteq\; A \times B$$

If there is a more abstract way of describing this, we can use set comprehension instead. Observing that the second element is always the square of the first:

$$S \;=\; \{\,(a,b) \;\mid\; (a,b) \in A \times B,\ b = a^2\,\}$$

So relations are typically defined as subsets of the Cartesian product, and we can either write the subset out explicitly or express it implicitly using set comprehension.

3 Two Relations on Numbers

Divisibility

Divisibility is an important relation on the natural numbers and integers. Recall that $d \mid n$ means that dividing n by d leaves remainder 0 — d divides n evenly.

$$D \;=\; \{\,(d,n) \;\mid\; (d,n) \in \mathbb{N} \times \mathbb{N},\ d \mid n\,\}$$

Pairs in the divisibility relation

$(7,63)$ because $7 \times 9 = 63$ $\qquad$ $(17,85)$ because $17 \times 5 = 85$

Careful with the bar

The vertical bar in $d \mid n$ is an arithmetic statement about numbers. It is *not* the same bar we use in set comprehension, which merely separates the parts of the notation.

This is a relation on pairs of natural numbers, so we obtain only the positive divisors. If we extend the generating set from $\mathbb{N}$ to $\mathbb{Z}$, we also pick up the negative divisors — since $(-7) \times (-9) = 63$, the two negative signs cancelling — and so we get a larger set of divisor pairs, with plus and minus versions of the pairs we had before.

Prime powers

A **prime power** is a prime multiplied by itself a certain number of times.

Some prime powers

$$5^2 = 25, \qquad 5^3 = 125, \qquad 5^4 = 625, \qquad 5^5 = 3125$$

$$343 \;=\; 7^3 \qquad \text{— also a prime power}$$

And $(3, 1)$ belongs to the relation because anything to the power 0 is 1 by definition: $3^0 = 1$. In fact, for every number, the pair with 1 is a prime power.

To define prime powers it is useful to define the primes first. One way is to say that the set of factors of the number consists of exactly two elements, 1 and the number itself:

$$P \;=\; \{\, p \;\mid\; p \in \mathbb{N},\ \text{factors}(p) = \{1, p\},\ p \neq 1 \,\}$$

Why the condition $p \neq 1$

Since sets do not distinguish duplicates, writing $\text{factors}(p) = \{1, p\}$ on its own would also admit $p = 1$: the factors of 1 are $\{1, 1\}$, which is just $\{1\}$. As we do not want to count 1 as a prime, we add the condition $p \neq 1$.

Now the prime powers are

$$\{\, (p, n) \;\mid\; (p, n) \in P \times \mathbb{N},\ n = p^m \text{ for some } m \in \mathbb{N} \,\}$$

Since m may be 0, this is exactly why $(3, 1)$ is included.

The language of comprehensions is flexible

We may write phrases like "for some" in a comprehension, or use mathematical symbols for "there exists" and "for all". Either is acceptable: we do not have to be rigid about the notation, as long as there is no ambiguity about what is meant.

4 A Relation Beyond Numbers: Airline Routes

Why are relations so interesting in the context of computing and data? Let us leave numbers behind.

Direct flights

Let C be the set of cities in which an airline operates. The airline flies directly between some pairs of cities but not all; for other pairs one must take a hopping flight from A to B and then

from B to C. The direct flights form a relation

$$D \subseteq C \times C.$$

Information about an airline's routes is therefore a relation in exactly the sense we mean.

Is it reflexive?

Since this is a relation between a set and itself, we can ask whether every element is related to itself.

Reflexive and irreflexive

reflexive: $(a, a) \in D$ for every a

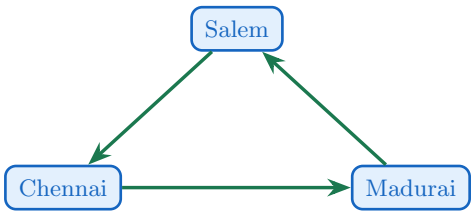
irreflexive: $(a, a) \notin D$ for every a

It is easy to see that direct flights should not be reflexive: we do not expect an airline to operate a flight that takes off from an airport and lands immediately back at the same airport. In fact we would like it to be precisely **irreflexive** — no airport should serve itself.

Is it symmetric?

Symmetry would say that whenever a pair of cities is in the relation, the reverse pair is too: if we can fly directly from one city to another, we can fly directly back.

For big cities this is usually the case — certainly for the metros and largest state capitals. But for smaller cities it need not hold. It is quite common for airlines to serve three cities on a **triangular route**: there may be a flight from Chennai to Madurai, but to come back from Madurai one may have to fly to Salem first.



no direct flight back from Madurai to Chennai

Between these three cities one can get from any to any other, either directly or indirectly, depending on the direction of travel. So this relation is irreflexive but **not necessarily symmetric** — it depends on the context.

5 Tables Are Relations

We can extend the route information into a **table** — useful for deriving other things, such as how long a flight takes or how expensive a ticket is likely to be. Here we record the flying distance between a pair of cities.

Source	Destination	Distance (km)
Bangalore	Chennai	290
Chennai	Delhi	1752
Delhi	Chennai	1752

A table is just a relation

Every column represents a potential set of values. The first column is a city, drawn from C; the second is also a city from C; the third is a natural number. So this table is a relation on

$$C \times C \times \mathbb{N}.$$

In general we have all possible pairs of cities and all possible numbers, but only some triples are interesting — namely those where the two cities are actually connected by a flight and the number is the real distance.

Properties of a relation are practical matters

Some entries are useless and need not be recorded even though we know them: for every city the flying distance to itself is 0, so there is no reason to store it.

More importantly, unlike the direct-flights relation, **distance is symmetric**. It does not matter whether there is a direct flight from Chennai to Delhi and back, or from Chennai to Madurai and not back: it is enough to record the distance once. As the table shows, Chennai to Delhi and Delhi to Chennai are both 1752 km, because that is how distances work.

Why knowing the property matters

When a relation is represented as a table, symmetry lets us save space by not recording the symmetric entries and noting separately that the relation is symmetric. So asking whether a relation is reflexive, irreflexive or symmetric is not merely an abstract question: a symmetric relation can be stored with only *half* the entries, the other half following by symmetry.

6 Keys

Tables also turn up when we look at data about people. A school or college would record information about students in this form:

Roll number	Name	Date of birth
21f1001	Abhay Kumar	12-03-2003
21f1002	Payal Ghosh	07-11-2002
21f1003	Jeremy Pinto	25-06-2003
21f1004	Payal Ghosh	19-01-2003

There would typically be further personal information as well — home address, phone number and so on.

Key

A column designed to be **unique**, so that no two rows share a value in it, is called a **key**. It lets us identify each student directly and individually, without confusion about which student we mean.

Some columns are not natural in the same way: everybody has a name and is born on a particular date, but the roll number is *assigned* by the school or college precisely so that it is unique. The other columns are not keys — two students may have the same name, and it is even possible for two students to share both a name and a date of birth. So we cannot rely on the other columns to distinguish rows.

A keyed table behaves like a function

If every student has a unique roll number, then each row is identified by that roll number: give the roll number, and the row — and hence all the other values — is determined. That is exactly what a function does: given an input, produce a unique output.
Such a stored table is sometimes called a set of **key–value pairs**. We can change the value for a given key by updating it; but adding a new entry requires a new key, so that there is no confusion.

7 The Join

A school usually maintains more than one table. There might be a separate table recording the grades of students in the courses they take. For conciseness we keep only the roll numbers and subject names there, not the student names.

Roll number	Subject	Grade
21f1002	Mathematics	A
21f1004	Physics	B
21f1001	Mathematics	B
21f1003	Physics	A

Now consider a typical requirement: generating a **report card**. The grade table has the roll number, subject and grade, but does not tell us who the student is — and apart from the students themselves, nobody would recognise which roll number belongs to whom.

We therefore want a table with the roll number, an extra column with the name taken from the first table, and then the subject and grade:

Roll number	Name	Subject	Grade
21f1002	Payal Ghosh	Mathematics	A
21f1004	Payal Ghosh	Physics	B
21f1001	Abhay Kumar	Mathematics	B
21f1003	Jeremy Pinto	Physics	A

Why keys matter here

The name *Payal Ghosh* is ambiguous — there are two of them — and they have two separate rows because they have two different roll numbers. The Payal Ghosh who got an A in mathematics is not the same as the Payal Ghosh who got a B in physics.

This operation combines two tables — and since a table is a relation, it combines two relations. It is an important operation in computing and data science called a **Join**.

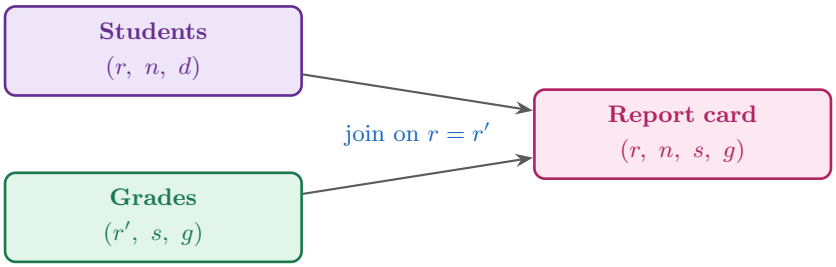
Join

A join takes tuples from two relations and combines them on **common values**:

$$\{\,(r, n, s, g) \mid (r, n, d) \in \text{Students},\ (r', s, g) \in \text{Grades},\ r = r'\,\}$$

We take any roll number, name and date of birth from Students, and any roll number, subject and grade from Grades, and require that the roll numbers on the two sides be the same. When

they are, we emit a new tuple keeping the name n from the left, the subject s and grade g from the right, and the common roll number r — throwing away the date of birth, which we are not interested in preserving.



This ensures that rows are never merged across two different students: the grade for Abhay will not be joined to the name and date of birth of Jeremy Pinto, because they have different roll numbers.

Tables are relations

A relation describes special tuples in a Cartesian product. From a computing and data science point of view we work with tables all the time, and tables really are relations — which is why relations play such a central role. Combining information across tables is an operation on relations; the join described here is only one kind, and other operations appear in later courses.

Lecture 11 — A Closer Look at Functions

Ranges of values, local and global maxima and minima, and comparing growth rates

What this lecture covers

We now look more closely at functions, and in particular at functions over numbers. Three questions occupy us: what range of values can a function actually attain, where does it reach its smallest and largest values, and how fast does it grow compared with another function?

1 Recalling the Definitions

A function is a rule mapping inputs to outputs. For numbers, a function might take an input x and map it to x^2, which we can name:

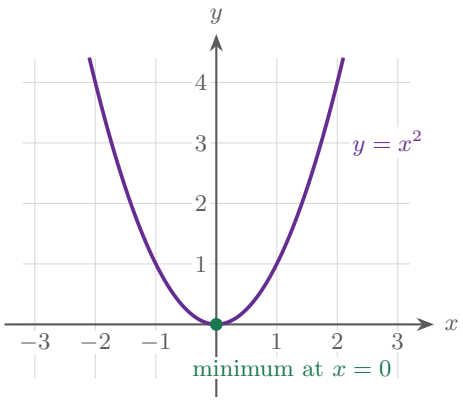
$$x \mapsto x^2 \qquad\qquad g(x) = x^2$$

With such a function come the three sets: the **domain**, the inputs that are allowed; the **codomain**, the set to which the outputs belong; and the **range**, the actual outputs this input set generates under the rule.

The associated relation is

$$\{\,(x,y) \mid x,y \in \mathbb{R},\ y = x^2\,\}$$

so the domain and codomain are both $\mathbb{R}$, and $y = x^2$ is the filter picking out the pairs we want. Plotting all the points of the relation gives the graph below, which shows that although the codomain is all of $\mathbb{R}$, the function never goes below 0: the **range** is the non-negative reals.



Functions need not be about numbers

We are not restricted to functions on numbers. Taking the set of all people — in the world, in a country, in any geographical region — we can look at the function **mother**, which maps a person uniquely to that person's mother. This is a function because every person has exactly one mother. In this course, however, we shall mostly study functions on numbers.

2 The Range of Values

One of the basic questions is: given a codomain, what range of values can we actually achieve through the function?

Example 1 — the square function

$$f(x) = x^2$$

This is always non-negative, so the outputs lie between 0 and $+\infty$. There is no upper bound, but we never obtain a negative value.

Example 2 — a cubic

$$f(x) = x^3 - 3x^2 + 5$$

When x becomes very small (large and negative), x^3 becomes very large in negative magnitude, since the cube of a negative number is negative: for instance

$$(-1000)^3 = (-1000) \times (-1000) \times (-1000) = -10^9.$$

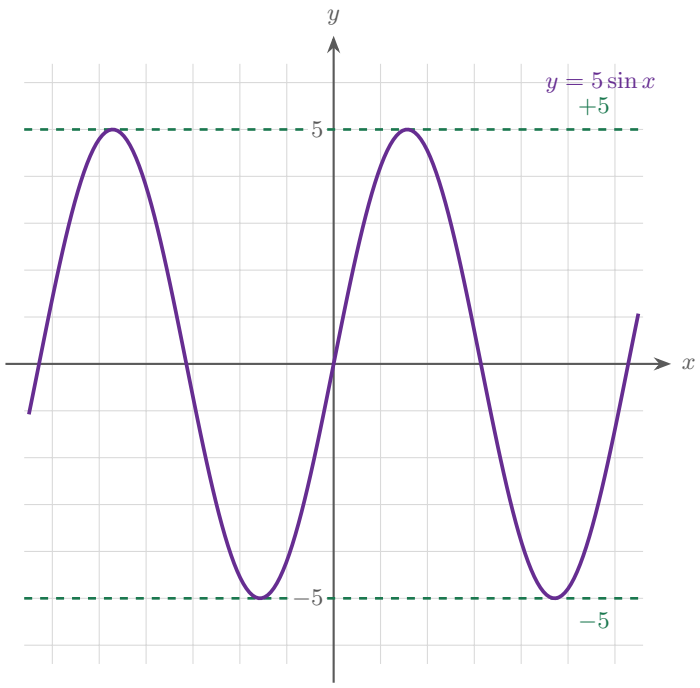
The same happens for large positive values in the positive direction. So this function has a range from $-\infty$ to $+\infty$.

Example 3 — a bounded range

Trigonometric functions such as $\sin x$ oscillate between an upper and a lower bound: $\sin x$ lies between -1 and $+1$, so

$$f(x) = 5\sin x \qquad \text{lies between } -5 \text{ and } +5.$$

Even considering all possible inputs, we never go outside this range.



3 Maxima and Minima

Within the range of values a function can take, we are often interested in specific points — in particular, where the values are smallest and where they are largest.

The square function

For $f(x) = x^2$ the graph makes it clear that at $x = 0$ the output is 0, and at every other point the output is bigger than 0. So the function attains its **minimum** value at $x = 0$. Since it keeps growing indefinitely on both sides, there is **no maximum**.

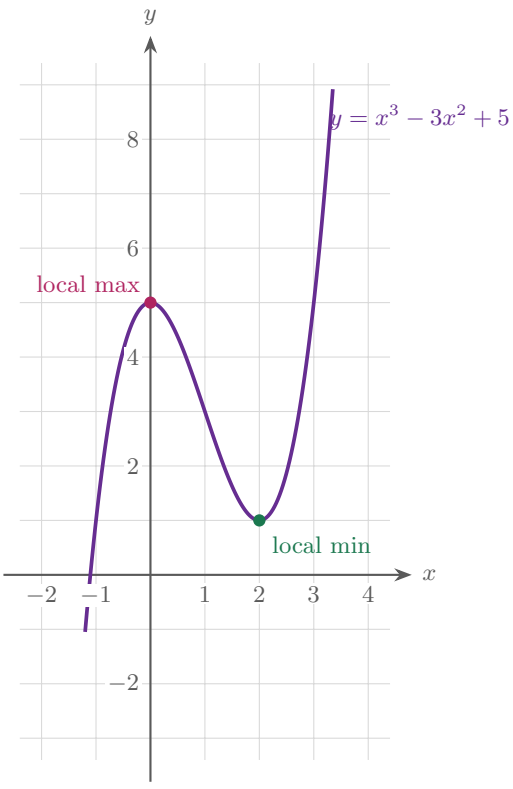
The cubic

The cubic $f(x) = x^3 - 3x^2 + 5$ grows arbitrarily small as we go to negative inputs and arbitrarily large in the positive direction, so it has **no maximum and no minimum** overall. But its behaviour in between is interesting, because it zigzags: it goes up, comes down, and goes up again.

at $x = 0$: it turns around — a **local maximum** — and starts falling

at $x = 2$: it turns around again — a **local minimum** — and rises

We are interested, for various reasons, in finding where these local maxima and minima are.



The bounded case

For $5 \sin x$, the value -5 is a local minimum and $+5$ a local maximum. Here they are also the **global** minimum and maximum, since these are the extreme values the function can ever attain — and they are attained infinitely often, periodically, as we move from left to right.

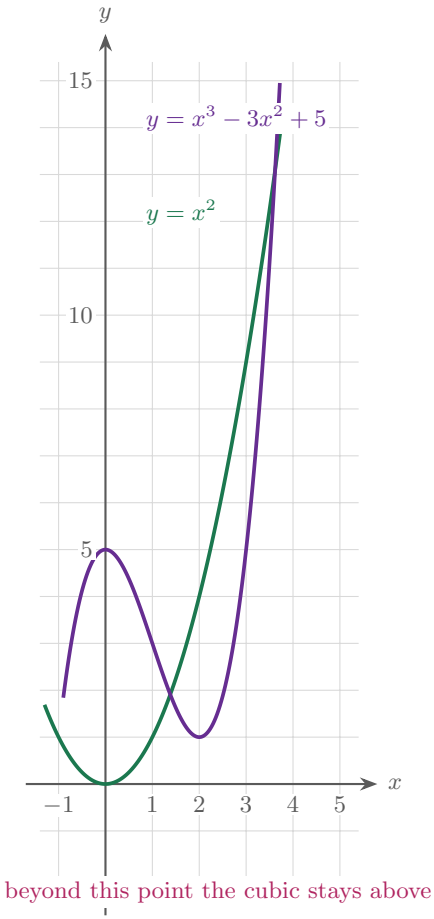
4 Growth Rates

Another thing we care about is **how fast** a function grows: does one function grow faster than another?

Comparing our two functions

$$y = x^2 \qquad \text{and} \qquad y = x^3 - 3x^2 + 5,$$

we see from the graphs that although the cubic is initially below the parabola in places, it eventually overtakes it and, after that point, is never below it again. In this sense the **cubic grows faster** than the square function.



Why growth rates matter

We often see growth compared informally in everyday contexts. Here is one relevant to you.

Graduates and jobs

Let

$$G(y) = \text{the number of data science students graduating in year } y,$$

$$J(y) = \text{the number of new data science jobs in year } y.$$

As the year increases — 2020, 2021, and so on — G takes some value, and hopefully this number is increasing because courses are growing. At the same time jobs are being created in data science.

Ideally the two are comparable. If the number of jobs increases faster than the number of graduates, there is demand for graduates, and more students will opt to study data science — so one would expect demand for such a course. The unfortunate case may of course happen the other way round: if demand slumps, people graduating with a degree in data science will not be employable, and there will be a reverse trend.

So when we look at data we are interested in comparing the growth rates of functions, and we shall study this in the context of the functions we treat mathematically.

What we shall ask of a function

Studying functions over numbers, several properties interest us: the range of outputs, where the function attains its local minima and local maxima, and the relative growth rates of different functions — along with many others we shall meet as we go along.

References and Acknowledgement

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